

Appendix A

Jacobi polynomials

Definition

$$P_n^{(\alpha, \beta)}(x) = \frac{(\alpha + 1)_n}{(n!)} \sum_{m=0}^n \frac{1}{m!} \frac{(-n)_m (n + \alpha + \beta + 1)_m}{(\alpha + 1)_m} \left(\frac{1-x}{2} \right)^m \quad (\text{B.1})$$

Differential equation

$$\begin{aligned} (1-x^2) \frac{\partial^2}{\partial x^2} P_n^{(\alpha, \beta)}(x) + (\beta - \alpha - (\alpha + \beta + 2)x) \frac{\partial}{\partial x} P_n^{(\alpha, \beta)}(x) \\ + n(n + \alpha + \beta + 1) P_n^{(\alpha, \beta)}(x) = 0 \end{aligned} \quad (\text{B.2})$$

Rodrigues formula

$$P_n^{(\alpha, \beta)}(x) = \frac{(-1)^n}{2^n n!} \frac{1}{(1-x)^\alpha (1+x)^\beta} \left(\frac{d}{dx} \right)^n ((1-x)^{n+\alpha} (1+x)^{n+\beta}). \quad (\text{B.3})$$

Orthogonality condition

$$\begin{aligned} \int_{-1}^1 (1-x)^\alpha (1+x)^\beta P_n^{(\alpha, \beta)}(x) P_m^{(\alpha, \beta)}(x) dx \\ = \frac{2^{\alpha+\beta+1}}{n!} \frac{\Gamma(n+\alpha+1) \Gamma(n+\beta+1)}{(2n+\alpha+\beta+1) \Gamma(n+\alpha+\beta+1)} \delta_{nm}. \end{aligned} \quad (\text{B.4})$$

Important functional relations

$$\begin{aligned}
& 2(n+1)(n+\alpha+\beta+1)(2n+\alpha+\beta)P_{n+1}^{(\alpha,\beta)}(x) = (2n+\alpha+\beta+1) \\
& \times [(2n+\alpha+\beta)(2n+\alpha+\beta+2)x + \alpha^2 - \beta^2]P_n^{(\alpha,\beta)}(x) \\
& - 2(n+\alpha)(n+\beta)(2n+\alpha+\beta+2)P_{n-1}^{(\alpha,\beta)}(x)
\end{aligned} \tag{B.5}$$

$$(2n+\alpha+\beta)P_n^{(\alpha-1,\beta)}(x) = (n+\alpha+\beta)P_n^{(\alpha,\beta)}(x) - (n+\beta)P_{n-1}^{(\alpha,\beta)}(x) \tag{B.6}$$

$$(2n+\alpha+\beta)P_n^{(\alpha,\beta-1)}(x) = (n+\alpha+\beta)P_n^{(\alpha,\beta)}(x) - (n+\alpha)P_{n-1}^{(\alpha,\beta)}(x) \tag{B.7}$$

$$\frac{d^m}{dx^m}[P_n^{(\alpha,\beta)}(x)] = \frac{1}{2^m} \frac{\Gamma(n+m+\alpha+\beta+1)}{\Gamma(n+\alpha+\beta+1)} P_{n-m}^{(\alpha+m,\beta+m)}(x). \tag{B.8}$$

Connections with hypergeometric functions

$$\begin{aligned}
P_n^{(\alpha,\beta)}(x) &= \frac{(-1)^n \Gamma(n+1+\beta)}{n! \Gamma(1+\beta)} F\left(n+\alpha+\beta+1, -n; 1+\beta; \frac{1+x}{2}\right); \\
&= \frac{\Gamma(n+1+\alpha)}{n! \Gamma(1+\alpha)} F\left(n+\alpha+\beta+1, -n; 1+\alpha; \frac{1-x}{2}\right); \\
&= \frac{\Gamma(n+1+\alpha)}{n! \Gamma(1+\alpha)} \left(\frac{1+x}{2}\right)^n F\left(-n, -n-\beta; 1+\alpha; \frac{x-1}{x+1}\right); \\
&= \frac{\Gamma(n+1+\beta)}{n! \Gamma(1+\beta)} \left(\frac{x-1}{2}\right)^n F\left(-n, -n-\alpha; 1+\beta; \frac{x+1}{x-1}\right).
\end{aligned} \tag{B.9}$$

Appendix B

Laguerre polynomials

Definition

$$L_n^{(\alpha)}(x) = \frac{1}{n!} \sum_{m=0}^n \frac{(-n)_m}{m!} (\alpha + m + 1)_{n-m} x^m \left(\frac{1-x}{2}\right)^k \quad (\text{C.1})$$

Differential equation

$$x \frac{\partial}{\partial x^2} L_n^{(\alpha)}(x) + (\alpha + 1 - x) \frac{\partial}{\partial x} L_n^{(\alpha)}(x) + n L_n^{(\alpha)}(x) = 0 \quad (\text{C.2})$$

Rodrigues formula

$$L_n^\alpha(x) = \frac{1}{n!} \frac{1}{e^{-x} x^\alpha} \left(\frac{d}{dx} \right)^n (e^{-x} x^{n+\alpha}). \quad (\text{C.3})$$

Orthogonality condition

$$\int_0^\infty e^{-x} x^\alpha L_n^{(\alpha)}(x) L_m^{(\alpha)}(x) dx = \frac{1}{n!} \Gamma(n + \alpha + 1) \delta_{nm}. \quad (\text{C.4})$$

Important functional relations

$$\frac{d}{dx} L_n^{(\alpha)}(x) = -L_{n-1}^{(\alpha+1)}(x) = \frac{1}{x} (n L_n^{(\alpha)}(x) - (n + \alpha) L_{n-1}^{(\alpha)}(x)). \quad (\text{C.5})$$

$$(n + 1) L_{n+1}^{(\alpha)}(x) - (2n + \alpha + 1 - x) L_n^{(\alpha)}(x) + (n + \alpha) L_{n-1}^{(\alpha)}(x) = 0 \quad (\text{C.6})$$

$$L_n^{(\alpha)}(x) - L_n^{(\alpha-1)}(x) = L_{n-1}^{(\alpha)}(x) \quad (\text{C.7})$$

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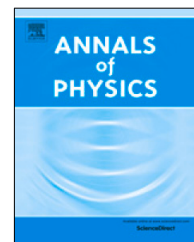
1. **Supersymmetry and Shape Invariance of Exceptional Orthogonal Polynomials**
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Supersymmetry and shape invariance of exceptional orthogonal polynomials

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Shape invariance

ABSTRACT

We discuss the exceptional Laguerre and the exceptional Jacobi orthogonal polynomials in the framework of the supersymmetric quantum mechanics (SUSYQM). We express the differential equations for the Jacobi and the Laguerre exceptional orthogonal polynomials (EOP) as the eigenvalue equations and make an analogy with the time independent Schrödinger equation to define “Hamiltonians” enables us to study the EOPs in the framework of the SUSYQM and to realize the underlying shape invariance associated with such systems. We show that the underlying shape invariance symmetry is responsible for the solubility of the differential equations associated with these polynomials.

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1. Introduction

In 2009 new families of orthogonal polynomials (known as exceptional orthogonal polynomials (EOP)), related to some of the old classical orthogonal polynomials are discovered [1–3]. Unlike the usual classical orthogonal polynomials, these EOPs start with degree $n = 1$ or higher integer values and still form a complete orthonormal set with respect to a positive definite inner product defined over a compact interval. Two of the well known classical orthogonal polynomials, namely the Laguerre orthogonal polynomials and the Jacobi orthogonal polynomials have been extended to the EOPs category. By X_m Laguerre (Jacobi) EOP means the complete set of Laguerre (Jacobi) orthogonal polynomials with degree $\geq m$. Here m is a positive integer and can have values 1, 2, 3 ... Soon after

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this remarkable discovery, the connection of EOPs with the translationally shape invariant potential was established [4–8]. The list of exactly solvable quantum mechanical potentials thus got enlarged with the corresponding energy eigen functions for the newly obtained exactly solvable systems being expressed in terms of EOPs. Such systems are known as rational extension of the original systems. These developments have greatly helped in extending the list of the exactly solvable potentials over the past decade [9–36]. Attempts have also been made to extend the classical Hermite polynomials to the EOP category [37].

In the present article we would like to study EOPs in the framework of the SUSYQM. The differential equations for the Jacobi and the Laguerre EOPs can be written as eigenvalue equations. We can associate a “Hamiltonian”¹ operator to each of the EOPs by comparing the associated differential equation with the time independent Schrödinger equation, $H\psi = E\psi$. Further, using the factorization properties of the EOPs [1,38] we show that these Hamiltonians are factorizable in terms of the raising and the lowering operators. These raising (lowering) operators when act on the EOPs, increase (decrease) the order of the polynomials by one unit. This enables us to construct the supersymmetric partner Hamiltonians for both the EOPs. Supersymmetric techniques are then used to get the solutions corresponding to these EOPs. Further by introducing an arbitrary integer parameter, we construct a sequence of supersymmetric partner Hamiltonians to show the underlying shape invariance associated with such systems. For, simplicity we first present the case of X_1 EOPs and later generalize to arbitrary m . It is worth adding that a similar study has already been undertaken for the classical orthogonal polynomials [39–41] as well as for hypergeometric function [42].

The plan of the paper is as follows. In Section 2 we briefly introduce the basic ideas of SUSYQM and SI which will be necessary for the analysis of the EOPs. In Section 3 we discuss X_1 Jacobi and the X_1 Laguerre EOPs within the framework of SUSYQM and bring out the SI associated with these EOPs which is responsible for the solubility of these EOPs. In Section 4 we generalize these results to the X_m EOPs. Finally we summarize the results in Section 5.

2. Supersymmetry and shape invariance

In this section we briefly review some key features of SUSYQM and shape invariance [43,44]. One dimensional Hamiltonian (in units of $2m = 1$, $\hbar = 1$) is factorized in terms of first order operators as

$$\begin{aligned} A &= \frac{d}{dx} + W(x) \\ A^\dagger &= -\frac{d}{dx} + W(x) \end{aligned} \quad (1)$$

where $W(x)$, a real function of x , is known as the super potential while $A^\dagger$ and A are raising and lowering operators, analogous to case of simple harmonic oscillator. The supersymmetric partner Hamiltonians are then defined as

$$\begin{aligned} H_- &= A^\dagger A = -\frac{d^2}{dx^2} + V_-(x) \\ H_+ &= A A^\dagger = -\frac{d^2}{dx^2} + V_+(x) \end{aligned} \quad (2)$$

where $V_\pm = W^2 \mp W'(x)$. These two Hamiltonians $H_\mp$ have exactly same spectrum except the lowest energy state of H_- , which is zero, i.e.

$$E_0^{(-)} = 0, \quad E_n^{(+)} = E_{n+1}^{(-)}. \quad (3)$$

¹ We would like to clarify that this “Hamiltonian” is not a true Hamiltonian which represents total energy of a classical system. Rather it is analogous to the true Hamiltonian. Therefore the eigenvalues of this analogous Hamiltonian do not represent the energy values, even though the eigenvalues are denoted by the symbols E_n . Such notations are adopted to make the clear analogy with SUSYQM techniques based on which the manuscript is prepared.

The eigenfunctions of $H_{\pm}$, $\psi_{\pm}^n$, are interrelated through the operations of $A^{\dagger}$, A . In particular,

$$\psi_{+}^n = A^{\dagger} \psi_{-}^{n+1}, \quad \psi_{-}^{n+1} = A \psi_{+}^n, \quad \psi_{-}^0 = N e^{\int^x W(y) dy}. \quad (4)$$

Thus SUSYQM relates the bound state eigenvalues as well as the corresponding eigen functions of the partner Hamiltonians without actually finding them.

This analysis can also be extended to the non-Hermitian systems [45]. In the case of non hermitian Hamiltonians the raising and lowering operators are not hermitian conjugate to each other and hence the supersymmetry partner Hamiltonians are like, BA and AB (instead of $A^{\dagger}A$ and $AA^{\dagger}$). In this case the spectrum needs not be bounded from below.

However, in case the two partner potentials are shape invariant, i.e. if they satisfy

$$V_{+}(x, a_0) = V_{-}(x, a_1) + R(a_1), \quad (5)$$

where a_1 is some function of a_0 , i.e. $a_1 = f(a_0)$ and $R(a_1)$ is a constant then one can show that the energy eigenvalues corresponding to $V_{+}(x)$ are given by

$$E_{n+1}^{(-)} = \sum_{i=1}^n R(a_i), \quad E_0^{(-)} = 0. \quad (6)$$

If there is a family of p shape invariant Hamiltonians, one can relate the p th eigenfunction of original Hamiltonian H^1 to H^p as

$$\psi_p^1(x; a_1) \propto B(x; a_1)B(x; a_2) \dots B(x; a_p)\psi_0^1(x; a_{p+1}) \quad (7)$$

where $B(x; a_p)$'s are the raising operators. These concepts will be used in later sections to realize the underlying symmetries associated with the EOPs.

3. X_1 -Exceptional orthogonal polynomials

In this section, we discuss the X_1 Jacobi and the X_1 Laguerre polynomials within the framework of SUSYQM and SI techniques and bring out the reasoning for the solubility of the corresponding X_1 Jacobi and the X_1 Laguerre differential equations.

3.1. X_1 -Jacobi orthogonal polynomials

The X_1 -Jacobi orthogonal polynomials $\hat{P}_n^{(\alpha, \beta)}(x)$ satisfy the differential equation

$$\begin{aligned} \hat{P}_n^{(\alpha, \beta)}(x) + \left(\frac{(\beta + \alpha + 2)x - (\beta - \alpha)}{x^2 - 1} + \frac{2(\beta - \alpha)}{(\beta + \alpha) - (\beta - \alpha)x} \right) \hat{P}_n^{(\alpha, \beta)}(x) \\ + \left(\frac{(\beta - \alpha)x - (n - 1)(n + \alpha + \beta)}{x^2 - 1} + \frac{(\beta - \alpha)^2}{(\beta + \alpha) - (\beta - \alpha)x} \right) \hat{P}_n^{(\alpha, \beta)}(x) = 0 \end{aligned} \quad (8)$$

where $n = 1, 2, 3, \dots$ and the real parameter $\alpha > -1$ and $\beta > -1$.

Eq. (8) can be expressed as an eigenvalue equation

$$\begin{aligned} \left[(x^2 - 1) \frac{d^2}{dx^2} + \left((\beta + \alpha + 2)x - (\beta - \alpha) + \frac{2(\beta - \alpha)(x^2 - 1)}{(\beta + \alpha) - (\beta - \alpha)x} \right) \frac{d}{dx} \right. \\ \left. + \left((\beta - \alpha)x + \frac{(\beta - \alpha)^2(x^2 - 1)}{(\beta + \alpha) - (\beta - \alpha)x} - (n - 1)(n + \alpha + \beta) \right) \right] \hat{P}_n^{(\alpha, \beta)}(x) = 0 \\ n = 1, 2, 3, \dots \end{aligned} \quad (9)$$

This can be formally put in the form

$$[H - (n - 1)(n + \alpha + \beta)] \hat{P}_n^{(\alpha, \beta)}(x) = 0 \quad (10)$$

where

$$H = (x^2 - 1) \frac{d^2}{dx^2} + \left((\beta + \alpha + 2)x - (\beta - \alpha) + \frac{2(\beta - \alpha)(x^2 - 1)}{(\beta + \alpha) - (\beta - \alpha)x} \right) \frac{d}{dx} + \left((\beta - \alpha)x + \frac{(\beta - \alpha)^2(x^2 - 1)}{(\beta + \alpha) - (\beta - \alpha)x} \right) \quad (11)$$

The raising and lowering operators for the X_1 -Jacobi polynomials are

$$A_{\alpha,\beta} = \frac{(x - c)^2}{x - b} \frac{d}{dx} \left(\frac{1}{x - c} \right) \quad (12a)$$

$$B_{\alpha,\beta} = (x^2 - 1) \frac{(x - b)}{(x - c)} \left(\frac{d}{dx} + a \right) - a(x^2 - 2bx + 1) \quad (12b)$$

with a , b and c being related to α , β by the following relations.

$$a = \frac{\beta - \alpha}{2} \quad (13a)$$

$$b = \frac{\beta + \alpha}{\beta - \alpha} \quad (13b)$$

$$c = b = \frac{1}{a} \quad (13c)$$

These raising and lowering operators hold the following relations

$$A_{\alpha,\beta} \hat{P}_n^{(\alpha,\beta)}(x) = \left(\frac{n + \alpha + \beta}{2} \right) \hat{P}_{n-1}^{(\alpha+1,\beta+1)}, \quad (14)$$

$$B_{\alpha,\beta} \hat{P}_n^{(\alpha+1,\beta+1)} = 2n \hat{P}_{n+1}^{(\alpha,\beta)}, \quad (15)$$

and the Hamiltonian in Eq. (11) can then be factorized in terms of these two operators as

$$H = B_{\alpha,\beta} A_{\alpha,\beta} = A_{\alpha-1,\beta-1} B_{\alpha-1,\beta-1} - (\alpha + \beta) \quad (16)$$

By using Eqs. (14) and (15) in Eq. (10) we have

$$B_{\alpha,\beta} A_{\alpha,\beta} \hat{P}_n^{(\alpha,\beta)} = E_n \hat{P}_n^{(\alpha,\beta)} \quad (17)$$

$$A_{\alpha,\beta} B_{\alpha,\beta} (A_{\alpha,\beta} \hat{P}_n^{(\alpha,\beta)}) = E_n (A_{\alpha,\beta} \hat{P}_n^{(\alpha,\beta)})$$

with

$$E_n = (n - 1)(n + \alpha + \beta) \quad (18)$$

Thus H^1 and H^2 as defined by

$$H^1 = B_{\alpha,\beta} A_{\alpha,\beta} \quad (19a)$$

$$H^2 = A_{\alpha,\beta} B_{\alpha,\beta} \quad (19b)$$

are supersymmetric partner Hamiltonians.

We can generalize this further and show that the pair of Hamiltonians $B_{\alpha+k-1,\beta+k-1} A_{\alpha+k-1,\beta+k-1}$ and $A_{\alpha+k-1,\beta+k-1} B_{\alpha+k-1,\beta+k-1}$ are the supersymmetric partner Hamiltonians for $k = 1, 2, \dots$. In particular, by using Eqs. (14) and (15) we can write,

$$B_{\alpha+k,\beta+k} A_{\alpha+k,\beta+k} \hat{P}_n^{(\alpha+k,\beta+k)} = E_n^k \hat{P}_n^{(\alpha+k,\beta+k)}, \quad (20)$$

and

$$A_{\alpha+k-1,\beta+k-1} B_{\alpha+k-1,\beta+k-1} \hat{P}_n^{(\alpha+k,\beta+k)} = E_{n+1}^{k-1} \hat{P}_n^{(\alpha+k,\beta+k)}, \quad (21)$$

where

$$E_n^{k-1} = (n - 1)(n + \alpha + \beta + 2k - 2). \quad (22)$$

Above equations imply

$$\begin{aligned} B_{\alpha+k,\beta+k} A_{\alpha+k,\beta+k} - E_n^k \\ = A_{\alpha+k-1,\beta+k-1} B_{\alpha+k-1,\beta+k-1} - E_{n+1}^{k-1}, \end{aligned} \quad (23)$$

where $(k = 1, 2, 3, \dots)$. From Eq. (23) it is clear that the supersymmetric pair of Hamiltonians $B_{\alpha+k-1,\beta+k-1} A_{\alpha+k-1,\beta+k-1}$ and $A_{\alpha+k-1,\beta+k-1} B_{\alpha+k-1,\beta+k-1}$ are in fact shape invariant with only a shift.

By using the above generalized operators we can find the X_1 Jacobi polynomials

$$2^{(s-1)}(s-1)! \hat{P}_s^{(\alpha,\beta)} = B_{\alpha,\beta} B_{\alpha+1,\beta+1} \dots B_{\alpha+s-2,\beta+s-2} \hat{P}_1^{(\alpha+s-1,\beta+s-1)} \quad (24)$$

where $s = 2, 3, 4, \dots$. Let us consider,

$$H^k = B_{\alpha+k-1,\beta+k-1} A_{\alpha+k-1,\beta+k-1} + (k-1)(k+\alpha+\beta)$$

It is straight forward to see that this sequence of Hamiltonians satisfy Shape invariant property and can be written as

$$= H^{k+1} = H^k + (\alpha + \beta + 2k) = H^k + R_k \quad (25)$$

where $R_k = (\alpha + \beta + 2k)$

So the n th eigenvalue of the original Hamiltonian (H^1), as expected, is given by

$$\begin{aligned} E_n &= \sum_{k=1}^{n-1} (\alpha + \beta) + 2k \\ &= (n-1)(n + \alpha + \beta). \end{aligned} \quad (26)$$

3.2. X_1 Laguerre polynomials

For $n \geq 1$ and the real parameter $k > 0$ the X_1 Laguerre polynomials $\hat{L}_n^{(k)}$ satisfy the differential equation

$$-x \hat{L}_n^{(k)''}(x) + \frac{(x-k)}{x+k} [(x+k+1) \hat{L}_n^{(k)'}(x)] - \left[\frac{(x-k)}{(x+k)} + (n-1) \right] \hat{L}_n^{(k)} = 0 \quad (27)$$

This differential equation can be written as the eigenvalue equation as

$$\left(-x \frac{d^2}{dx^2} + \frac{(x-k)}{(x+k)} \left[(x+k+1) \frac{d}{dx} - 1 \right] - (n-1) \right) \hat{L}_n^{(k)} = 0, \quad (28)$$

which can be compactly expressed as

$$[H - (n-1)] \hat{L}_n^{(k)} = 0, \quad (29)$$

where

$$H = -x \frac{d^2}{dx^2} + \frac{(x-k)}{(x+k)} \left[(x+k+1) \frac{d}{dx} - 1 \right]. \quad (30)$$

One can define the raising and the lowering operators for the X_1 Laguerre polynomials as

$$A_k = -\frac{(x+k+1)^2}{x+k} \frac{d}{dx} \left(\frac{1}{x+k+1} \right), \quad (31a)$$

$$B_k = \frac{x(x+k)}{(x+k+1)} \left(\frac{d}{dx} - 1 \right) + k \quad (31b)$$

The operators A_k and B_k satisfy following raising and lowering properties

$$A_k \hat{L}_n^{(k)} = \hat{L}_{n-1}^{(k+1)}, \quad (32a)$$

$$B_k \hat{L}_n^{(k+1)} = n \hat{L}_{n+1}^{(k)}. \quad (32b)$$

Hence the Hamiltonian in Eq. (30) can be factorized in terms of these two operators as

$$H = B_k A_k = A_{k-1} B_{k-1} - 1. \quad (33)$$

Using Eqs. (32a) and (32b) we obtain

$$\begin{aligned} B_k A_k \hat{L}_n^{(k)} &= E_n^0 \hat{L}_n^{(k)}, \\ A_k B_k (A_k \hat{L}_n^{(k)}) &= E_n^0 (A_k \hat{L}_n^{(k)}) \text{ or } A_k B_k (\hat{L}_{n-1}^{(k+1)}) = E_n^0 (\hat{L}_{n-1}^{(k+1)}), \end{aligned} \quad (34)$$

where,

$$E_n^0 = (n - 1).$$

This indicates that the Hamiltonians $A_k B_k$ and $B_k A_k$ are supersymmetric partner Hamiltonians. Now to establish the shape invariance, we construct the sequence of Hamiltonians $A_{k+r-1} B_{k+r-1}$ and $B_{k+r-1} A_{k+r-1}$ with $r = 1, 2, 3, \dots$. It is straightforward to check

$$\begin{aligned} B_{k+r-1} A_{k+r-1} \hat{L}_n^{(k+r-1)} &= E_n^{r-1} \hat{L}_n^{(k+r-1)}, \\ A_{k+r-1} B_{k+r-1} (A_{k+r-1} \hat{L}_n^{(k+r-1)}) &= E_n^{r-1} (A_{k+r-1} \hat{L}_n^{(k+r-1)}) \text{ or } A_{k+r-1} B_{k+r-1} \hat{L}_{n-1}^{(k+r)} = E_n^{r-1} \hat{L}_{n-1}^{(k+r)} \end{aligned} \quad (35)$$

with $E_n^{r-1} = (n - 1)$. From the recurrence relations we can write

$$\begin{aligned} B_{k+r} A_{k+r} - (n - 1) \\ = A_{k+r-1} B_{k+r-1} - n. \end{aligned} \quad (36)$$

The above relation shows that the supersymmetric partner Hamiltonians in Eq. (35) are also shape invariant with a constant shift, independent of r .

We can also relate the X_1 Laguerre polynomials by using raising operators in the following way

$$(s - 1)! \hat{L}_s^{(k)} = B_k B_{k+1} B_{k+2} \dots B_{k+s-2} \hat{L}_1^{(k+s-1)} \quad (37)$$

with $s = 2, 3, 4, \dots$. Let us define

$$H^r = B_{k+r-1} A_{k+r-1} + (r - 1). \quad (38)$$

We can see that the sequence of shape invariant Hamiltonians satisfy

$$\begin{aligned} A_{k+r-1} B_{k+r-1} + (r - 1) \\ = B_{k+r} A_{k+r} + 1 + r - 1 \\ = B_{k+r} A_{k+r} + r \\ = H^{r+1} = H^r + 1. \end{aligned} \quad (39)$$

Hence the n th eigenvalue of the Hamiltonian H^1 is

$$\begin{aligned} E_n &= \sum_{k=1}^{n-1} 1 \\ &= (n - 1). \end{aligned} \quad (40)$$

4. X_m -Orthogonal polynomials

The results obtained in the last section are easily generalized for the X_m EOPs as we show now.

4.1. X_m -Jacobi orthogonal polynomials

The X_m Jacobi orthogonal polynomials $\hat{P}_{n,m}^{(\alpha,\beta)}(x)$ satisfy the differential equation

$$\begin{aligned} \hat{P}_{n,m}^{(\alpha,\beta)}(x) + \left[(\alpha - \beta - m + 1) \frac{P_{m-1}^{(-\alpha,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} - \left(\frac{\alpha+1}{1-x} \right) + \left(\frac{\beta+1}{1+x} \right) \right] \\ * \hat{P}_{n,m}^{(\alpha,\beta)}(x) + \frac{1}{(1-x^2)} \left[\beta(\alpha - \beta - m + 1)(1-x) \frac{P_{m-1}^{(-\alpha,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} \right. \\ \left. + m(\alpha - \beta - m + 1) + (n-m)(\alpha + \beta + n - m + 1) \right] \hat{P}_{n,m}^{(\alpha,\beta)}(x) = 0 \end{aligned} \quad (41)$$

Where $P_m^{(\alpha,\beta)}(x)$ are the classical Jacobi polynomials.

The X_m Jacobi differential equation can be written as the eigenvalue equation

$$\begin{aligned} \left[(x^2 - 1) \frac{d^2}{dx^2} + (x^2 - 1) \left((\alpha - \beta - m + 1) \frac{P_{m-1}^{(-\alpha,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} - \left(\frac{\alpha+1}{1-x} \right) + \left(\frac{\beta+1}{1+x} \right) \right) \frac{d}{dx} \right. \\ \left. + \left(\beta(\alpha - \beta - m + 1)(x-1) \frac{P_{m-1}^{(-\alpha,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} - m(\alpha - \beta - m + 1) - (n-m)(\alpha + \beta + n - m + 1) \right) \right] \hat{P}_{n,m}^{(\alpha,\beta)}(x) = 0 \end{aligned} \quad (42)$$

This can be formally written as

$$[H - (n-m)(\alpha + \beta + n - m + 1)] \hat{P}_{n,m}^{(\alpha,\beta)}(x) = 0, \quad (43)$$

where

$$\begin{aligned} H = (x^2 - 1) \frac{d^2}{dx^2} + (x^2 - 1) \left((\alpha - \beta - m + 1) \frac{P_{m-1}^{(-\alpha,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} - \left(\frac{\alpha+1}{1-x} \right) + \left(\frac{\beta+1}{1+x} \right) \right) * \frac{d}{dx} \\ + \beta(\alpha - \beta - m + 1)(x-1) \frac{P_{m-1}^{(-\alpha,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} - m(\alpha - \beta - m + 1) \end{aligned} \quad (44)$$

The lowering and raising operators for the X_m Jacobi polynomials are

$$A_{\alpha,\beta}^m = \frac{P_m^{(-\alpha-2,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} \left[\frac{d}{dx} - \frac{1}{2}(\beta - \alpha + m - 1) \frac{P_{m-1}^{(-\alpha-1,\beta+1)}(x)}{P_m^{(-\alpha-2,\beta)}(x)} \right], \quad (45)$$

and

$$\begin{aligned} B_{\alpha,\beta}^m = (1-x^2) \frac{P_m^{(-\alpha-1,\beta-1)}(x)}{P_m^{(-\alpha-2,\beta)}(x)} \left[\frac{d}{dx} - \left(\frac{1}{2}(\beta - \alpha + m - 1) \frac{P_{m-1}^{(-\alpha,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} \right. \right. \\ \left. \left. + \left(\frac{\alpha+1}{1-x} \right) - \left(\frac{\beta+1}{1+x} \right) \right) \right] \end{aligned} \quad (46)$$

These lowering and raising operators hold the following relations

$$A_{\alpha,\beta}^m \hat{P}_{n,m}^{(\alpha,\beta)}(x) = \frac{1}{2}(\alpha + \beta + n - m + 1) \hat{P}_{n-1,m}^{(\alpha+1,\beta+1)}(x) \quad n \geq m \quad (47)$$

and

$$B_{\alpha,\beta}^m \hat{P}_{n,m}^{(\alpha+1,\beta+1)}(x) = 2(n-m+1) \hat{P}_{n+1,m}^{(\alpha,\beta)}(x) \quad n \geq m. \quad (48)$$

Now by using the above recurrence relations, the Hamiltonian in (44) can be factorized as

$$H = B_{\alpha,\beta}^m A_{\alpha,\beta}^m = A_{\alpha-1,\beta-1}^m B_{\alpha-1,\beta-1}^m - (\alpha + \beta). \quad (49)$$

On using Eqs. (47) and (48), we can write

$$\begin{aligned} B_{\alpha,\beta}^m A_{\alpha,\beta}^m \hat{P}_{n,m}^{(\alpha,\beta)}(x) &= E_{n,m}^0 \hat{P}_{n,m}^{(\alpha,\beta)}(x), \\ A_{\alpha,\beta}^m B_{\alpha,\beta}^m (A_{\alpha,\beta}^m \hat{P}_{n,m}^{(\alpha,\beta)}(x)) &= E_{n,m}^0 (A_{\alpha,\beta}^m \hat{P}_{n,m}^{(\alpha,\beta)}(x)), \end{aligned} \quad (50)$$

where

$$E_{n,m}^0 = (n - m)(\alpha + \beta + n - m + 1). \quad (51)$$

Thus the Hamiltonians

$$\begin{aligned} H^1 &= B_{\alpha,\beta}^m A_{\alpha,\beta}^m, \\ H^2 &= A_{\alpha,\beta}^m B_{\alpha,\beta}^m, \end{aligned} \quad (52)$$

are supersymmetric partners. Now we construct a sequence of Hamiltonians by introducing a parameter k and find that the pair of Hamiltonians $B_{\alpha+k-1,\beta+k-1}^m A_{\alpha+k-1,\beta+k-1}^m$ and $A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m$ are also supersymmetric partner Hamiltonians for $k = 1, 2, 3, \dots$

$$\begin{aligned} B_{\alpha+k-1,\beta+k-1}^m A_{\alpha+k-1,\beta+k-1}^m P_{n,m}^{(\alpha+k-1,\beta+k-1)}(x) &= E_{n,m}^{k-1} P_{n,m}^{(\alpha+k-1,\beta+k-1)}(x) \\ A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m (A_{\alpha+k-1,\beta+k-1}^m P_{n,m}^{(\alpha+k-1,\beta+k-1)}(x)) &= E_{n,m}^{k-1} (A_{\alpha+k-1,\beta+k-1}^m P_{n,m}^{(\alpha+k-1,\beta+k-1)}(x)) \end{aligned} \quad (53)$$

where $k = 1, 2, 3, \dots$ and

$$E_{n,m}^{k-1} = (n - m)(\alpha + \beta + n - m + 2k - 1) \quad (54)$$

We can write

$$B_{\alpha+k,\beta+k}^m A_{\alpha+k,\beta+k}^m \hat{P}_{n,m}^{(\alpha+k,\beta+k)}(x) = E_{n,m}^k \hat{P}_{n,m}^{(\alpha+k,\beta+k)}(x), \quad (55)$$

and

$$A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m \hat{P}_{n,m}^{(\alpha+k,\beta+k)}(x) = E_{n+1,m}^{k-1} \hat{P}_{n,m}^{(\alpha+k,\beta+k)}(x). \quad (56)$$

From Eqs. (55) and (56) we can write,

$$B_{\alpha+k,\beta+k}^m A_{\alpha+k,\beta+k}^m = A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m + E_{n,m}^k - E_{n+1,m}^{k-1}, \quad (57)$$

where ($k=1,2,3, \dots$). Now from Eq. (57) it is clear that the supersymmetric pairs of Hamiltonians in Eq. (53) are in fact shape invariant with only a shift.

By using the above generalized operators we can write the X_m Jacobi polynomials as

$$2^{(s-m)}(s - m)! \hat{P}_{s,m}^{(\alpha,\beta)}(x) = B_{\alpha,\beta}^m B_{\alpha+1,\beta+1}^m \cdots B_{\alpha+s-m-1,\beta+s-m-1}^m \hat{P}_{m,m}^{(\alpha+s-m,\beta+s-m)}(x) \quad (58)$$

Where $s = m + 1, m + 2, \dots$. To see that, let us define a family of Hamiltonians

$$H^k = B_{\alpha+k-1,\beta+k-1}^m A_{\alpha+k-1,\beta+k-1}^m + (k - 1)(k + \alpha + \beta). \quad (59)$$

We can see that these shape invariant Hamiltonians satisfy

$$\begin{aligned} A_{\alpha+k-1,\beta+k-1}^m B_{\alpha+k-1,\beta+k-1}^m + (k - 1)(k + \alpha + \beta) \\ = B_{\alpha+k,\beta+k}^m A_{\alpha+k,\beta+k}^m + (\alpha + \beta + 2k) + (k - 1)(k + \alpha + \beta). \end{aligned} \quad (60)$$

Hence, we can write

$$H^{k+1} = H^k + R_k \quad (61)$$

where $R_k = (\alpha + \beta + 2k)$

So the n th eigenvalue of Hamiltonian H^1 is given by

$$\begin{aligned} E_n^m &= \sum_{k=1}^{n-m} (\alpha + \beta) + 2k \\ &= (n-m)(\alpha + \beta + n - m + 1). \end{aligned} \quad (62)$$

4.2. X_m -Laguerre orthogonal polynomials

For an integer $m \geq 0$, the X_m Laguerre orthogonal polynomials $\hat{L}_{n,m}^k(x)$ satisfy the differential equation

$$\begin{aligned} \hat{L}_{n,m}^k(x) + \frac{1}{x} \left((k+1-x) - 2x \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} \right) \hat{L}_{n,m}^k(x) \\ + \frac{1}{x} \left(n - 2k \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} \right) \hat{L}_{n,m}^k(x) = 0, \end{aligned} \quad (63)$$

where L_m^k are the classical Laguerre polynomials.

The above differential equation can be written in the form of eigenvalue equation as

$$\begin{aligned} \left[-x \frac{d^2}{dx^2} + \left((x-k-1) + 2x \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} \right) \frac{d}{dx} \right. \\ \left. + \left(2k \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} - n \right) \right] \hat{L}_{n,m}^k(x) = 0. \end{aligned} \quad (64)$$

This can be formally put in the form

$$[H - (n-m)] \hat{L}_{n,m}^k(x) = 0, \quad (65)$$

where

$$H = -x \frac{d^2}{dx^2} + \left((x-k-1) + 2x \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} \right) \frac{d}{dx} + \left(2k \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} - m \right). \quad (66)$$

The lowering and raising operators for the X_m Laguerre orthogonal polynomials are

$$A_k^m = -\frac{L_m^k(-x)}{L_m^{k-1}(-x)} \left(\frac{d}{dx} - \frac{L_{m-1}^{k+1}(-x)}{L_m^k(-x)} \right), \quad (67)$$

and

$$B_k^m = \frac{L_m^{k-1}(-x)}{L_m^k(-x)} \left(x \frac{d}{dx} + (1+k) \right) - x. \quad (68)$$

These operators hold the following relations

$$A_k^m \hat{L}_{n,m}^k(x) = \hat{L}_{n-1,m}^{k+1}(x) \quad (69)$$

and

$$B_k^m \hat{L}_{n,m}^{k+1}(x) = (n-m+1) \hat{L}_{n+1,m}^k(x) \quad (70)$$

From the above two recurrence relations we can write the Hamiltonian in Eq. (66) in a factorized form as

$$H = B_k^m A_k^m = A_{k-1}^m B_{k-1}^m - 1. \quad (71)$$

On using Eqs. (69) and (70) we can write

$$\begin{aligned} B_k^m A_k^m \hat{L}_{n,m}^k(x) &= E_{n,m}^0 \hat{L}_{n,m}^k(x) \\ A_k^m B_k^m (A_k^m \hat{L}_{n,m}^k(x)) &= E_{n,m}^0 (A_k^m \hat{L}_{n,m}^k(x)), \end{aligned} \quad (72)$$

where $E_{n,m}^0 = (n - m)$.

The above conditions show that the Hamiltonians $A_k^m B_k^m$ and $B_k^m A_k^m$ are supersymmetric partner Hamiltonians.

Now we introduce a parameter r and find that the pairs of Hamiltonians $A_{k+r-1}^m B_{k+r-1}^m$ and $B_{k+r-1}^m A_{k+r-1}^m$ are supersymmetric partners, for $r = 1, 2, 3 \dots$, by satisfying the conditions

$$\begin{aligned} B_{k+r-1}^m A_{k+r-1}^m \hat{L}_{n,m}^{k+r-1}(x) &= E_{n,m}^{r-1} \hat{L}_{n,m}^{k+r-1}(x) \\ A_{k+r-1}^m B_{k+r-1}^m (A_{k+r-1}^m \hat{L}_{n,m}^{k+r-1}(x)) &= E_{n,m}^{r-1} (A_{k+r-1}^m \hat{L}_{n,m}^{k+r-1}(x)), \end{aligned} \quad (73)$$

where $E_{n,m}^{r-1} = (n - m)$.

We see that the eigenvalues of the Hamiltonians are independent of r . So all n eigenstates have the degenerate eigenvalues.

From the recurrence relation we can write

$$\begin{aligned} B_{k+r}^m A_{k+r}^m - (n - m) \\ = A_{k+r-1}^m B_{k+r-1}^m - (n - m + 1). \end{aligned} \quad (74)$$

The above relation shows that the supersymmetric pair of Hamiltonians in Eq. (73) are in fact shape invariant with a constant shift.

We can also relate the X_m Laguerre polynomials by using raising operators in the following way

$$(s - m)! \hat{L}_{s,m}^{(k)} = B_k^m B_{k+1}^m B_{k+2}^m \dots B_{k+s-m-1}^m \hat{L}_{m,m}^{(k+s-m)} \quad (75)$$

where $s = m, m + 1, \dots$. Let us define

$$H^r = B_{k+r-1}^m A_{k+r-1}^m + (r - 1) \quad (76)$$

we can see that the sequence of shape invariant Hamiltonian satisfy

$$\begin{aligned} A_{k+r-1}^m B_{k+r-1}^m + (r - 1) \\ = B_{k+r}^m A_{k+r}^m + r, \end{aligned} \quad (77)$$

or in other words $H^{r+1} = H^r + 1$ so that the n 'th eigen value of the Hamiltonian is

$$E_n^m = \sum_{k=1}^{n-m} 1 = (n - m). \quad (78)$$

5. Conclusion

Exceptional Laguerre and exceptional Jacobi polynomials are used extensively in the rational extension of many quantum mechanical problems and hence worth investigating them in detail. In this paper we have described a new way² of looking at the differential equations for these EOPs and their solutions in the framework of supersymmetry and shape invariance. Supersymmetry and Shape invariance of the ‘‘Hamiltonians’’ corresponding to X_1 -Jacobi and X_1 -Laguerre polynomials have explicitly been shown in this work. The underlying shape invariance symmetry is responsible for the solubility of differential equations associated with these EOPs. Further these results are then generalized for X_m -Jacobi and X_m Laguerre polynomials with $m = 1, 2, 3 \dots$. We would like to add that classical Legendre and hypergeometric polynomials have been analysed in this fashion in [41,42]. In the present work we have extended their work for the EOPs. In Ref. [46] some master conditions are provided to check this for classical polynomials. It will be interesting to investigate EOPs along the line of [46] to obtain similar master condition for EOPs.

² We would like to point out that the standard technique to reduce the differential equations for the EOPs into Schrödinger form do not work using change of variable in a simple way.

CRediT authorship contribution statement

Satish Yadav: Formal analysis, Methodology. **Avinash Khare:** Conceptualization, Writing – review & editing, Supervision. **Bhabani Prasad Mandal:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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New solutions of Isochronous potentials in terms of exceptional orthogonal polynomials in heterostructures

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Abstract:	Point canonical transformation (PCT) has been used to find out new exactly solvable potentials in the position-dependent mass (PDM) framework. We solve 1-D Schrödinger equation in the PDM framework by considering two different fairly generic position-dependent masses (i) $M(x) = \lambda g_0(x)$ and, with $\eta = 0, 1, 2, \dots$. In the first case, we find new exactly solvable potentials that depend on an integer parameter m , and the corresponding solutions are written in terms of X_m -Laguerre polynomials. In the latter case, we obtain a new one parameter (v) family of isochronous solvable potentials whose bound states are written in terms of X_m -Laguerre polynomials. Further, we show that the new potentials are shape invariant by using the supersymmetric approach in the framework of PDM.
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New solutions of Isochronous potentials in terms of exceptional orthogonal polynomials in heterostructures

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Abstract

Point canonical transformation (PCT) has been used to find out new exactly solvable potentials in the position-dependent mass (PDM) framework. We solve 1-D Schrödinger equation in the PDM framework by considering two different fairly generic position-dependent masses (i) $M(x) = \lambda g'(x)$ and (ii) $M(x) = c(g'(x))^\nu$, $\nu = \frac{2\eta}{2\eta+1}$, with $\eta = 0, 1, 2, \dots$. In the first case, we find new exactly solvable potentials that depend on an integer parameter m , and the corresponding solutions are written in terms of X_m -Laguerre polynomials. In the latter case, we obtain a new one parameter (ν) family of isochronous solvable potentials whose bound states are written in terms of X_m -Laguerre polynomials. Further, we show that the new potentials are shape invariant by using the supersymmetric approach in the framework of PDM.

1 Introduction

In quantum mechanics, the Schrödinger equation for a few systems is solved exactly though the traditional means [1], the factorization methods of supersymmetric quantum mechanics (SUSY) [2–7], Lie algebraic techniques [8–10], reduction approach (to a hypergeometric equation [11] or to a confluent Heun equation [12]), Nikiforov-Uvarov (NU) method [13, 14] etc. Searching for analytically solvable potentials, whose energy spectrum are completely known, constitute an important field to study the low dimensional structures e.g. quantum well, quantum dot [15], specially the linear and nonlinear optical properties of the same [16–18]. On the other hand, the scenario of non-constant mass particularly position-dependent mass (PDM), although initially pursued in condensed matter physics problems, has become an active area of research in many disciplines of physics. The significant advancement in crystallographic growth techniques that enable the manufacture of non-uniform semiconductor specimens with abrupt heterojunctions is what has drawn the attention to the PDM approach. This kind of spatial dependencies have provided useful insights into new classes of phenomena such as the transport proprieties in semiconductors [15, 19], compositionally graded crystals [20], quantum dots [21], liquid crystals [22], and extended systems governed by superintegrability [23, 24]. The concept of PDM is also used for generating the pseudo-potentials which have an essential computational advantage in quantum Monte Carlo method [25, 26].

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On aside earlier studies on fundamental length scale (to remove divergences in field theories) show that any modification to the underlying space or to the canonical commutation relations (i.e Heisenberg uncertainty relation) [27, 28] can result to a Schrödinger equation with a position-dependent mass [29, 30]. To study these inhomogeneous materials we need to have an effective way to solve the PDM Schrödinger equation (PDMSE) which we will address in this article. Before going further it needs to be emphasized that in any scheme for the same leads to non-commutativity of the momentum and mass-operators and it results in an ordering ambiguity in the kinetic energy (KE) representation (see, e.g., [31, 32]). To avoid this we will use the Ben-Daniel and Duke model of PDMSE [33] for our study.

The bound state solutions of the Schrödinger equation are generally associated with some classical orthogonal polynomials such as Hermite polynomials, Laguerre polynomials, Legendre polynomials, etc. In 2009, new families of orthogonal polynomials connected to their classical counterparts were introduced and termed exceptional orthogonal polynomials (EOPs) [34–36]. These polynomials form an orthogonal and complete set of polynomials with a positive weight function even if their sequence starts with a degree $n \geq 1$. After this remarkable discovery, the quantum mechanical exploration of exactly solvable systems has been boosted, particularly those whose solutions are written in terms of EOPs. These are known as rational extensions of usual quantum systems. Over the past decade, these advancements have significantly expanded the list of exactly solvable potentials. [37–44].

In this article, we consider the PCT approach to construct the new exactly solvable potentials in a PDM background. We study the two cases, in the first case, we consider $M(x) = \lambda g'(x)$ which was also considered [45] to obtain the solution of PDMSE in terms of X_1 -Laguerre polynomials. We find new exactly solvable potentials and the eigenstates associated with these potentials are written in terms of X_m -Laguerre EOPs. It is important to note that numerous research have already been done on the intriguing problem of how classical and quantum "generalised harmonicity" relate to one another to which end we encounter with isochronous potentials [46, 47]. Stillinger and Stillinger [46] have suggested a two-parameter family of isochronous potentials, interpolating between the harmonic oscillator and the asymmetric parabolic well, while Dorignac has explored the quantum spectrum of the same which comes as equispaced [47]. In the second case, we propose a general form of $M(x) = c(g'(x))^\nu$, $\nu = \frac{2\eta}{2\eta+1}$, with $\eta = 0, 1, 2, \dots$ and obtain a one-parameter family of isochronous potentials that are exactly solvable and asymmetric in nature. Solutions are constructed in terms of X_m -Laguerre EOPs. It is worth adding that a similar study has already been undertaken for different choices of $M(x)$ [45, 48, 49]. Furthermore, by using the supersymmetric approach in quantum mechanics in the context of a PDEM framework, we show that these exactly solvable potentials are shape-invariant.

The organization of this paper is as follows: in section 2 we briefly introduce the position-dependent mass Schrödinger equation and deformed algebra. In section 3 we have generated two new exactly solvable potentials by using the point canonical transformation approach in the position-dependent effective mass background whose wavefunctions are written in terms of X_m Laguerre orthogonal polynomials. In section 4 we have shown that these new exactly solvable potentials are shape invariant via using SUSY in quantum mechanics in PDM background. Finally, in section 5 we summarize the result.

2 PDM Schrödinger Equation and Deformed Algebra

To avoid the noncommutativity, we start with the von Roos prescription of the two-parameter formulation of an effective-mass kinetic energy (KE) operator [50], which has an inbuilt Hermiticity (which also yields different plausible special cases), reads as

$$\hat{T} = \frac{1}{4}[\mathcal{M}^r(x)p\mathcal{M}^s(x)p\mathcal{M}^t(x) + \mathcal{M}^t(x)p\mathcal{M}^s(x)p\mathcal{M}^r(x)] \quad (1)$$

where r , s and t constitute a set of ambiguity parameters which follow $r + s + t = -1$. Using the above $\hat{T}$, we can write the Hamiltonian for PDMSE as follows

$$H = \left[\frac{1}{4} [\mathcal{M}^r(x) p \mathcal{M}^s(x) p \mathcal{M}^t(x) + \mathcal{M}^t(x) p \mathcal{M}^s(x) p \mathcal{M}^r(x)] + V(x) \right], \quad (2)$$

where $V(x)$ is the external potential. Of the frequently used Hermitian Hamiltonians, a few possible choices of the ambiguity parameters, have been explored in the literature for that see [51] and references therein.

Adopting units such that $\hbar = 2\mathcal{M}_0 = 1$ the PDMSE corresponding to the Hamiltonian in (2) can be projected as

$$\left[-\frac{1}{2} \left(M^r(x) \frac{d}{dx} M^s(x) \frac{d}{dx} M^t(x) + M^t(x) \frac{d}{dx} M^s(x) \frac{d}{dx} M^r(x) \right) + V(x) \right] \psi(x) = E \psi(x) \quad (3)$$

where $M(x)$ is defined by $\mathcal{M}(x) = \mathcal{M}_0 M(x)$ and is dimensionless. On setting $M(x) = f^{-2}(x)$ and $f(x) = 1 + \mathfrak{g}(x)$ [52], where $f(x)$ is some positive-definite function and $\mathfrak{g}(x) = 0$ corresponds to the constant-mass case, equation (3) becomes

$$\left[-\frac{1}{2} \left(f^r(x) \frac{d}{dx} f^s(x) \frac{d}{dx} f^t(x) + f^t(x) \frac{d}{dx} f^s(x) \frac{d}{dx} f^r(x) \right) + V(x) \right] \psi(x) = E \psi(x) \quad (4)$$

with $r + s + t = 2$. We can get rid of the ambiguity parameters r , s , t (denoted collectively by a) in the kinetic energy term by transferring them to the effective potential energy of the variable-mass system. Then using the result

$$f^r \frac{d}{dx} f^s \frac{d}{dx} f^t + f^t \frac{d}{dx} f^s \frac{d}{dx} f^r = 2\sqrt{f} \frac{d}{dx} f \frac{d}{dx} \sqrt{f} - (1 - r - t) f f'' - 2\left(\frac{1}{2} - r\right)\left(\frac{1}{2} - t\right) f'^2 \quad (5)$$

where the prime denotes derivative with respect to x and the positive definiteness of f is explicitly assumed, equation (4) acquires the form

$$H\psi(x) = \left[-\left(\sqrt{f(x)} \frac{d}{dx} \sqrt{f(x)} \right)^2 + V_{eff}(x) \right] \psi(x) = E\psi(x) \quad (6)$$

in which the effective potential

$$V_{eff}(x) = V(x) + \frac{1}{2}(1 - r - t)f(x)f''(x) + \left(\frac{1}{2} - r\right)\left(\frac{1}{2} - t\right)f'^2(x) \quad (7)$$

The PDMSE (6) may now be reinterpreted as the deformed Schrödinger equation as given by

$$H\psi(x) = [\pi^2 + V_{eff}(x)] \psi(x) = E\psi(x) \quad (8)$$

where the standard momentum operator $p = -i \frac{d}{dx}$ is now modified by the following deformed quantity defined by $\pi = -i \sqrt{f(x)} \frac{d}{dx} \sqrt{f(x)}$ and we thus observe that with respect to π the conventional commutation relation $[x, p] = i\hbar$ is modified to $[x, \pi] = i\hbar f(x)$ where $f(x)$ acts as a deforming function [27, 52, 53]. This type of deformation also has been extensively studied in different areas of physics such as in statistical physics (known as q -deformed algebra) [54, 55], in optics [56, 57] as well as black hole physics [58, 59]. In the next section, we will use the PCT and EOPs to obtain a class of exactly solvable potentials and their eigen spectrum.

3 Generation of new potential by PCT approach

Following the procedure outlined in the previous section the PDMSE in Eq. (3) can be written as

$$H\psi(x) = \left[-\frac{d}{dx} \frac{1}{M(x)} \frac{d}{dx} + V_{eff} \right] \psi(x) = E\psi(x) \quad (9)$$

where the effective potential is

$$V_{eff}(x) = V(x) + \frac{s+1}{2} \frac{M''}{M^2} - [r(r+s+1) + s+1] \frac{M'^2}{M^3} \quad (10)$$

$V(x)$ is the applied potential in the system. We will follow the choice of BenDaniel and Duke [33], i.e. $r = t = 0$ and $s = -1$ for which the effective and applied potentials become identical. We use the solution $\psi(x)$ in the form given by [45, 49, 52]

$$\psi(x) = f(x)F(g(x)) \quad (11)$$

where $F(g(x))$ satisfies second order differential equation

$$\frac{d^2 F}{dg^2} + Q(g) \frac{dF}{dg} + R(g)F = 0 \quad (12)$$

Now substituting this $\psi(x)$ in Eq. (9) we get

$$\frac{d^2 F}{dg^2} + \left(\frac{g''}{g'^2} + \frac{2f'}{fg'} - \frac{M'}{Mg'} \right) \frac{dF}{dg} + \left(\frac{f''}{fg'^2} - \frac{M'f'}{Mfg'^2} + (E - V_{eff}) \frac{M}{g'^2} \right) F = 0 \quad (13)$$

On comparison we get

$$Q(g(x)) = \left(\frac{g''}{g'^2} + \frac{2f'}{fg'} - \frac{M'}{Mg'} \right) \quad (14)$$

$$R(g(x)) = \left(\frac{f''}{fg'^2} - \frac{M'f'}{Mfg'^2} + (E - V_{eff}) \frac{M}{g'^2} \right) \quad (15)$$

Eq. (14) admits the solution [49]

$$f(x) \propto \sqrt{\frac{M}{g'}} e^{\frac{1}{2} \int^{g(x)} Q(t) dt} \quad (16)$$

Putting it in (15) we get

$$E - V_{eff} = \frac{g'''}{2Mg'} - \frac{3}{4M} \left(\frac{g''}{g'} \right)^2 + \frac{g'^2}{M} \left(R - \frac{1}{2} \frac{dQ}{dg} - \frac{Q^2}{4} \right) - \frac{M''}{2M^2} + \frac{3M'^2}{4M^3} \quad (17)$$

To ensure the presence of an energy-like term on the right-hand side of the above equation, which will compensate for the term E on its left-hand side, we can introduce a suitable relation between $M(x)$ and $g(x)$. This will give rise to an effective potential $V_{eff}(x)$ and the solution of the corresponding Schrödinger equation will be well-behaved. Three different choices of $M(x) \propto (g'(x))^\mu$ with $\mu = -1, 1, 2$ have been used extensively in various applications of the PCT approach in the PDM context [48, 60]. In this article, we consider a new type of position-dependent mass

$$M(x) = c (g'(x))^\nu \quad (18)$$

where c is a proportionality constant, $\nu = \frac{2\eta}{2\eta+1}$ and $\eta = 0, 1, 2, \dots$ along with $M(x) = \lambda g'(x)$

3.1 Case 1

For choice of $M(x) = \lambda g'(x)$, Eq.(17) becomes

$$E - V_{eff} = \frac{g'}{\lambda} \left(R - \frac{1}{2} \frac{dQ}{dg} - \frac{Q^2}{4} \right). \quad (19)$$

We further consider the differential equation for X_m -Laguerre polynomials and compare it with equation (12) to get

$$Q = \frac{1}{g} \left((\alpha + 1 - g) - 2g \frac{L_{m-1}^\alpha(-g)}{L_m^{\alpha-1}(-g)} \right), \quad \text{and} \quad R = \frac{1}{g} \left(n - 2\alpha \frac{L_{m-1}^\alpha(-g)}{L_m^{\alpha-1}(-g)} \right). \quad (20)$$

where $F(g) \propto \hat{L}_{n,m}^{(\alpha)}(g)$.

For these values of $Q(g(x))$ and $R(g(x))$ equation (19) reads

$$\begin{aligned} E - V_{eff} = & \frac{g'}{\lambda g} \left(n - \left(m - \frac{1}{2} \right) + \frac{\alpha}{2} + \frac{m}{\alpha} \right) - \frac{g'}{4\lambda} - \frac{g'}{4g^2\lambda} (\alpha^2 - 1) - \frac{2g' L_m^\alpha(-g)^2}{\lambda (L_m^{\alpha-1}(-g))^2} \\ & + \frac{g' L_{m-2}^{\alpha+1}(-g)}{\lambda (L_m^{\alpha-1}(-g))} - \frac{g' L_{m-1}^\alpha(-g)}{\lambda (L_m^{\alpha-1}(-g))} - \frac{g' L_{m-1}^{\alpha+1}(-g)}{\alpha \lambda (L_m^{\alpha-1}(-g))} + \frac{g' L_{m-1}^{\alpha+1}(-g)}{\lambda (L_m^{\alpha-1}(-g))} \end{aligned} \quad (21)$$

We observe that choice $\frac{g'}{\lambda g} = C$ will provide a constant term on the right-hand side to compensate for the energy term on the left-hand side. To get increasing eigenvalues for successive n -values, C must be restricted to positive. The solution of the choice $\frac{g'}{\lambda g} = C$ leads

$$g(x) = e^{-bx} \quad \text{and} \quad M(x) = e^{-bx}, \quad -\infty < x < \infty \quad (22)$$

where we have considered $C = b^2$ and $C\lambda = -b$, $b > 0$.

Substituting Eq.(22) in (21) we obtain the effective potential and energy eigenvalues

$$\begin{aligned} V_{eff} = & \frac{b^2}{4} (e^{bx} (\alpha^2 - 1) + e^{-bx}) + b^2 \left(\frac{2(L_{m-1}^\alpha(-e^{-bx}))^2}{(L_m^{\alpha-1}(-e^{-bx}))^2} - \frac{L_{m-2}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})} \right. \\ & \left. + \frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{\alpha L_m^{\alpha-1}(-e^{-bx})} + \frac{L_{m-1}^\alpha(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})} - \frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})} \right) e^{-bx} + V_c^I \end{aligned} \quad (23)$$

$$E_n^m = b^2 \left(n + \frac{\alpha + 1}{2} + \frac{m}{\alpha} \right) + V_c^I, \quad n = 0, 1, 2, \dots \quad (24)$$

where V_c^I is an arbitrary constant. The eigenfunctions are obtained using Eqs.(11) and (16)

$$\psi_n^m(x) = \mathcal{N} \frac{\exp[-\frac{1}{2}((\alpha + 1)bx + e^{-bx})]}{L_m^{\alpha-1}(-e^{-bx})} \hat{L}_{n+m,m}^\alpha(e^{-bx}) \quad n = 0, 1, 2, \dots \quad (25)$$

where $\mathcal{N}$ is the normalization constant given by

$$\mathcal{N} = \left(\frac{b n!}{(n + m + \alpha) \Gamma(n + \alpha)} \right)^{\frac{1}{2}}$$

We would like to point out that $m = 1$ case was done in [45] our general result for X_m case reproduces the result of [45] as

$$V_{eff} = \frac{b^2}{4} \left(e^{bx} (\alpha^2 - 1) + e^{-bx} + \frac{4}{\alpha(1 + \alpha e^{bx})} + \frac{8}{(1 + \alpha e^{bx})^2} \right) + V_c^I \quad (26)$$

$$E_n^1 = b^2 \left(n + \frac{\alpha + 1}{2} + \frac{1}{\alpha} \right) + V_c^I, \quad n = 0, 1, 2, \dots \quad (27)$$

$$\psi_n^1(x) = \mathcal{N} \frac{\exp[-\frac{1}{2}((\alpha + 1)bx + e^{-bx})]}{\alpha + e^{-bx}} \hat{L}_{n+1}^\alpha(e^{-bx}) \quad n = 0, 1, 2, \dots \quad (28)$$

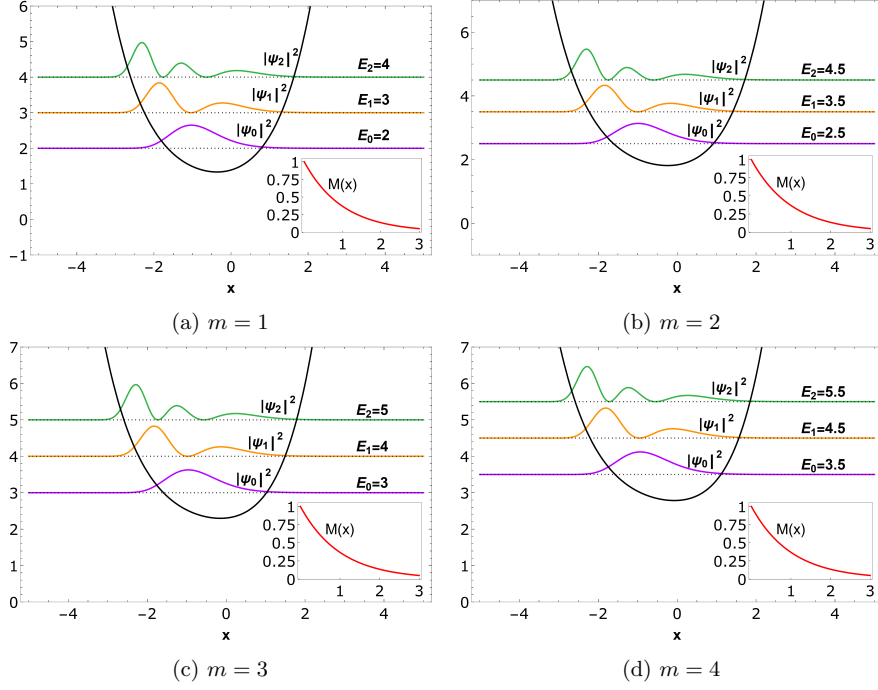


Figure 1: Plot of the potential V_{eff} given in Eq. (23) for different m values, square of first three bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), $|\psi_2|^2$ (green line) corresponding to different m values, for mass function $M(x)$ (red line) given in Eq. (22). We have consider here $b = 1$, $\alpha = 2$

3.2 Case 2

In this case, we consider a new type of position-dependent mass $M(x) = c(g'(x))^\nu$, $\nu = \frac{2\eta}{2\eta+1}$ and $\eta = 0, 1, 2, \dots$. For this choice Eq. (17) becomes

$$E - V_{eff} = \frac{g'''}{2cg'^{(\nu+1)}} - \frac{3g''^2}{4cg'^{(\nu+2)}} + \frac{1}{cg'^{(\nu-2)}} \left(R - \frac{1}{2} \frac{dQ}{dg} - \frac{Q^2}{4} \right) - \frac{(\nu^2 - \nu)g''^2}{2cg'^{(\nu+2)}} - \frac{\nu g'''}{2cg'^{(\nu+1)}} + \frac{3\nu^2 g''^2}{4cg'^{(\nu+2)}} \quad (29)$$

We further consider the differential equation for X_m -Laguerre polynomials with equation (12) to get

$$Q = \frac{1}{g} \left((\alpha + 1 - g) - 2g \frac{L_{m-1}^\alpha(-g)}{L_m^{\alpha-1}(-g)} \right), \quad \text{and} \quad R = \frac{1}{g} \left(n - 2\alpha \frac{L_{m-1}^\alpha(-g)}{L_m^{\alpha-1}(-g)} \right). \quad (30)$$

where $F(g) \propto \hat{L}_{n,m}^{(\alpha)}(g)$.

For these values of $Q(g(x))$ and $R(g(x))$ equation (29) becomes

$$\begin{aligned}
E - V_{eff} = & \frac{g'^{(2-\nu)}}{gc} \left(n - \left(m - \frac{1}{2} \right) + \frac{\alpha}{2} + \frac{m}{\alpha} \right) - \frac{g'^{(2-\nu)}}{4c} - \frac{g'^{(2-\nu)}}{4g^2c} (\alpha^2 - 1) - \frac{2g'^{(2-\nu)} L_m^\alpha(-g)^2}{c (L_m^{\alpha-1}(-g))^2} \\
& + \frac{g'^{(2-\nu)} L_{m-2}^{\alpha+1}(-g)}{c (L_m^{\alpha-1}(-g))} - \frac{g'^{(2-\nu)} L_{m-1}^\alpha(-g)}{c (L_m^{\alpha-1}(-g))} - \frac{g'^{(2-\nu)} L_{m-1}^{\alpha+1}(-g)}{\alpha c (L_m^{\alpha-1}(-g))} + \frac{g'^{(2-\nu)} L_{m-1}^{\alpha+1}(-g)}{c (L_m^{\alpha-1}(-g))} \\
& - \frac{(\nu^2 - \nu)g''^2}{2cg'^{(\nu+2)}} - \frac{\nu g'''}{2cg'^{(\nu+1)}} + \frac{3\nu^2 g''^2}{4cg'^{(\nu+2)}} + \frac{g'''}{2cg'^{(\nu+1)}} - \frac{3g''^2}{4cg'^{(\nu+2)}}
\end{aligned} \tag{31}$$

By following a similar approach as in the case 1, we observe that the choice $g'^{(2-\nu)} = \kappa c g$, where κ is a constant, will provide a constant term on the right-hand side of the equation, corresponding to the energy term E on the left-hand side. To get increasing eigenvalues for successive n values, it is necessary to restrict the constant κ to positive values. The solution of mentioned first order differential equation $g'^{(2-\nu)} = \kappa c g$ reads $g(x) = x^{\frac{2-\nu}{1-\nu}}$, which generate $M(x) = \left(\frac{2-\nu}{1-\nu} \right)^2 x^{\frac{\nu}{1-\nu}}$. For simplicity we may take $\kappa c = \left(\frac{2-\nu}{1-\nu} \right)^2$ such that $\kappa = 1$ and $c = \left(\frac{2-\nu}{1-\nu} \right)^2$. Substituting these values in Eq.(31) we obtain the effective potential and energy eigenvalues as

$$\begin{aligned}
V_{eff} = & \frac{1}{4} \left(x^{\frac{\nu-2}{1-\nu}} (\alpha^2 - 1) + x^{\frac{2-\nu}{1-\nu}} \right) + \left(\frac{2(L_{m-1}^\alpha(-x^{\frac{2-\nu}{1-\nu}}))^2}{(L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}}))^2} - \frac{L_{m-2}^{\alpha+1}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})} + \frac{L_{m-1}^{\alpha+1}(-x^{\frac{2-\nu}{1-\nu}})}{\alpha L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})} \right. \\
& \left. + \frac{L_{m-1}^\alpha(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})} - \frac{L_{m-1}^{\alpha+1}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})} \right) x^{\frac{2-\nu}{1-\nu}} + \frac{(1-\nu)(3-\nu)}{4(2-\nu)^2} x^{\frac{\nu-2}{1-\nu}} + V_c^{II}
\end{aligned} \tag{32}$$

$$E_n^m = \left(n + \frac{\alpha+1}{2} + \frac{m}{\alpha} \right) + V_c^{II}, \quad n = 0, 1, 2, \dots \tag{33}$$

where, V_c^{II} is arbitrary constant. The eigenfunctions are obtained using Eqs.(11) and (16)

$$\psi_n^m(x) = \mathcal{N} \frac{\exp\left(-\frac{x^{\frac{2-\nu}{1-\nu}}}{2}\right)}{x^{\frac{1}{2}} (L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}}))} x^{\frac{(2-\nu)(1+\alpha)}{2(1-\nu)}} \hat{L}_{n+m,m}^\alpha(x^{\frac{2-\nu}{1-\nu}}), \quad n = 0, 1, 2, \dots \tag{34}$$

where $\mathcal{N}$ is the normalization constant and is given by

$$\mathcal{N} = \left(\frac{n!}{(n+m+\alpha)\Gamma(n+\alpha)} \right)^{\frac{1}{2}}$$

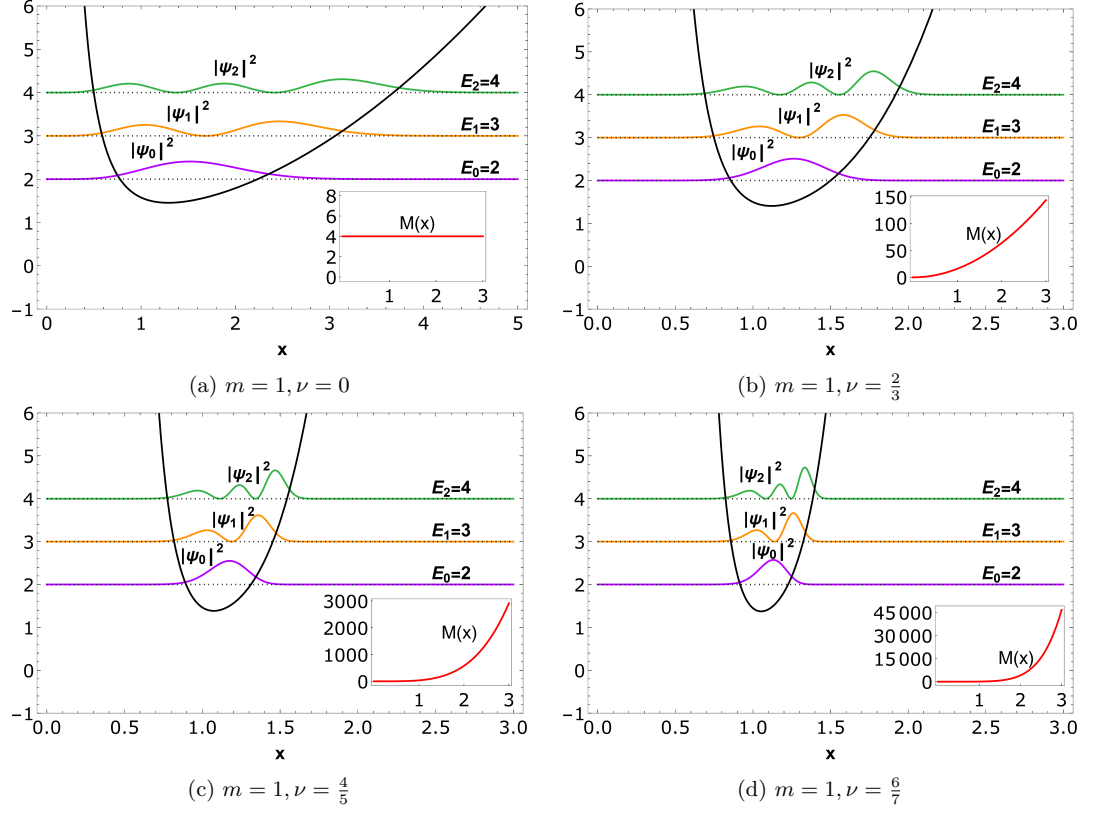


Figure 2: Plot of the potential V_{eff} given in Eq. (32) for $m = 1$ and different ν values, square of first three bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), $|\psi_2|^2$ (green line) corresponding to different ν values, for mass function $M(x)$ (red line). We have consider here $\alpha = 2$.

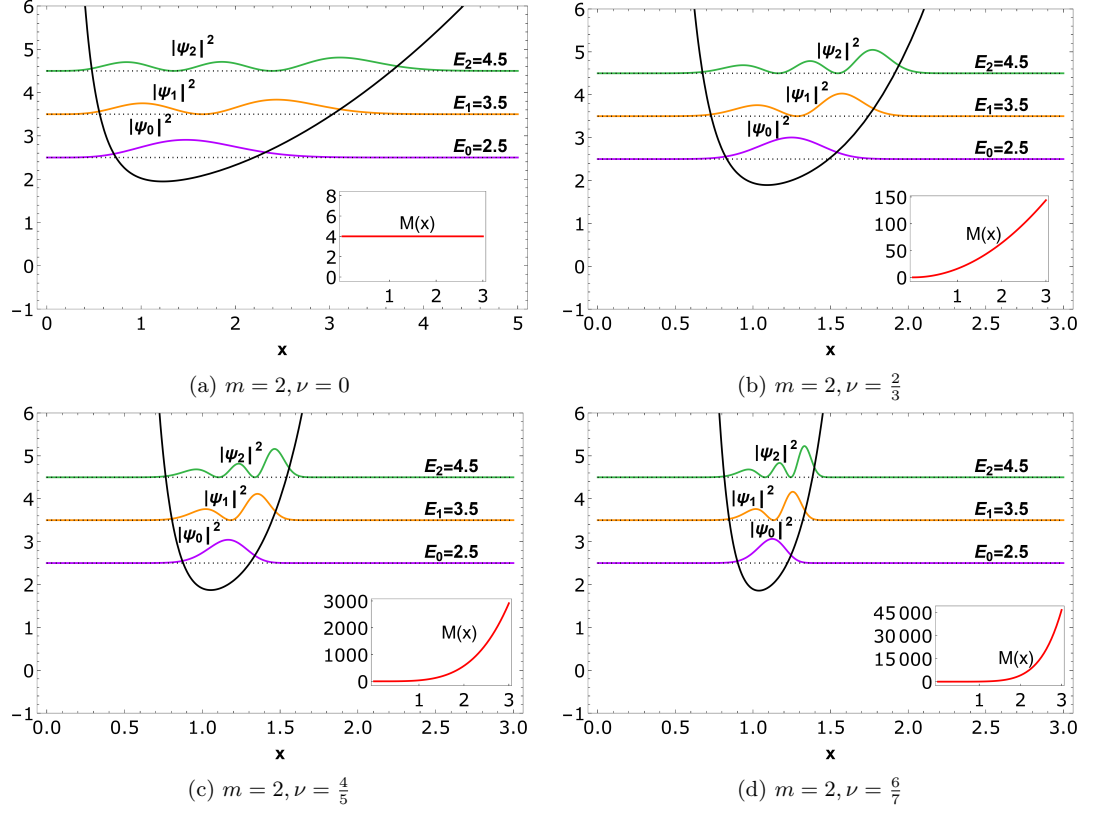


Figure 3: Plot of the potential V_{eff} given in Eq. (32) for $m = 2$ and different ν values, square of first three bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), $|\psi_2|^2$ (green line) corresponding to different ν values, for mass function $M(x)$ (red line). We have consider here $\alpha = 2$

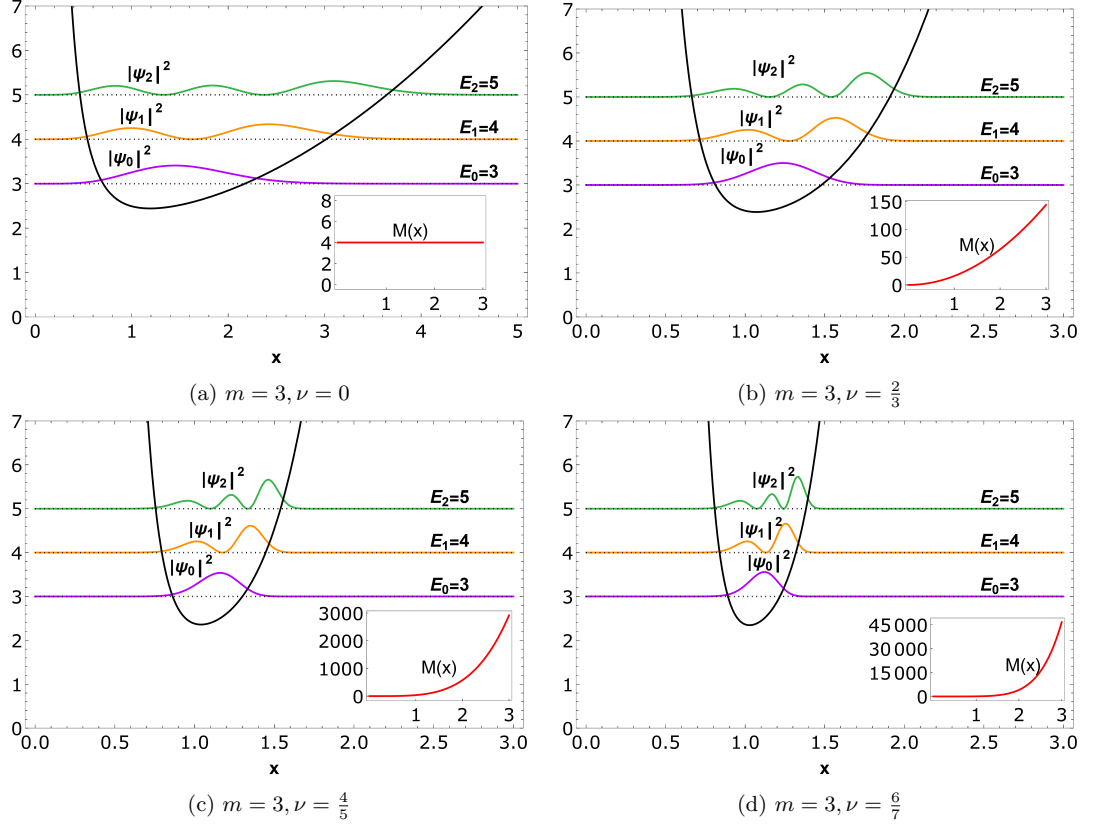


Figure 4: Plot of the potential V_{eff} given in Eq. (32) for $m = 3$ and different ν values, square of first three bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), $|\psi_2|^2$ (green line) corresponding to different ν values, for mass function $M(x)$ (red line). We have considered here $\alpha = 2$

We further generalise this analysis for the two dimensional case. The probability density of a two-dimensional potential is shown in the Fig. 5.

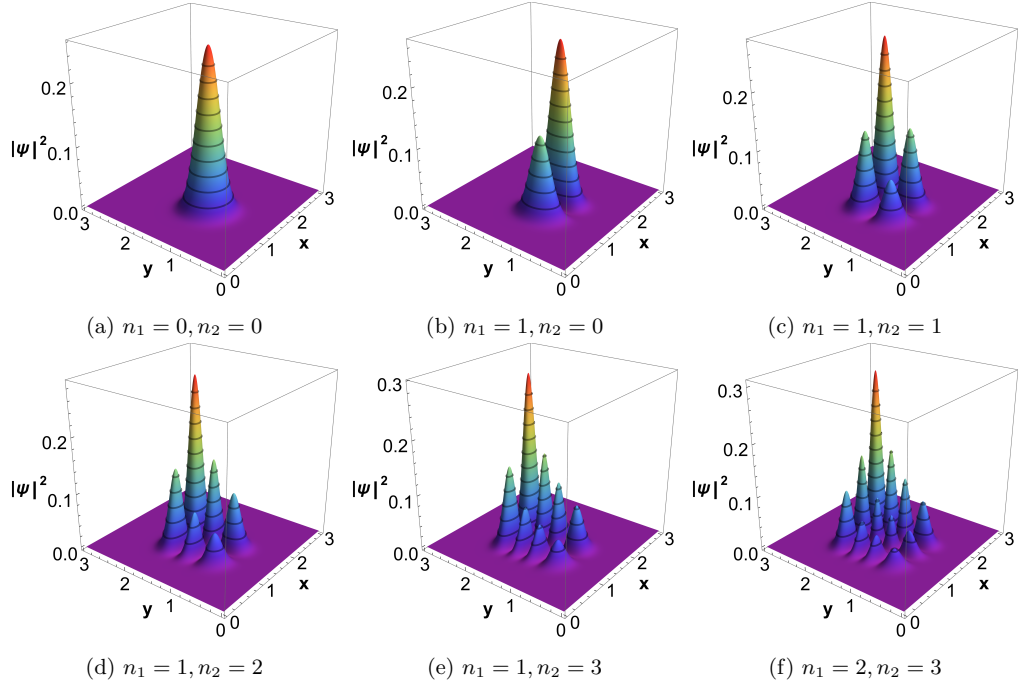


Figure 5: Plot for the probability density of a two-dimensional potential for $m = 1, \nu = \frac{2}{3}$ and different combinations of n_1 and n_2 values. The wavefunctions are normalized.

4 Supersymmetric approach

Within the context of supersymmetric quantum mechanics applied to systems with an effective mass, one can define the lowering and raising operator [61]

$$\mathcal{A}\psi = \frac{1}{\sqrt{M}} \frac{d\psi}{dx} + \mathcal{W}\psi, \quad \text{and} \quad \mathcal{A}^\dagger\psi = -\frac{d}{dx} \left(\frac{\psi}{\sqrt{M}} \right) + \mathcal{W}\psi \quad (35)$$

where $\mathcal{W} = -\frac{1}{\sqrt{M}} \frac{\psi_0'}{\psi_0}$ is the superpotential, and ψ_0 represents the ground state. The Hamiltonian for a system with an effective mass of (9) can be factorized as follows.

$$H_{eff} = \mathcal{A}^\dagger \mathcal{A} = -\frac{d}{dx} \left(\frac{1}{M(x)} \right) + V_{eff} \quad (36)$$

The associated supersymmetric partner Hamiltonian is given by

$$H_{eff}^p = \mathcal{A} \mathcal{A}^\dagger = -\frac{d}{dx} \left(\frac{1}{M(x)} \right) + V_{eff}^p \quad (37)$$

We see that both partner Hamiltonians H_{eff} and H_{eff}^p describe particles with identical spatial dependence on effective mass but in different potentials V_{eff} and V_{eff}^p respectively. These supersymmetric partner potentials are given by

$$V_{eff} = -\left(\frac{\mathcal{W}}{\sqrt{M}} \right)' + \mathcal{W}^2 \quad (38)$$

$$V_{eff}^p = V_{eff} + \frac{2\mathcal{W}'}{\sqrt{M}} - \left(\frac{1}{\sqrt{M}}\right) \left(\frac{1}{\sqrt{M}}\right)'' \quad (39)$$

If these potentials are shape invariant, they must satisfy the condition

$$V_{eff}^p(x, a_1) = V_{eff}(x, a_2) + R(a_1) \quad (40)$$

where a_1 is a set of parameters, a_2 is some function of a_1 and $R(a_1)$ is independent of x . For unbroken supersymmetry, the eigenvalues and the eigenfunctions of the two such Hamiltonians with position-dependent effective mass are related by ($n = 0, 1, 2, \dots$)

$$E_n^{eff,p} = E_{n+1}^{eff}, \quad E_0^{eff} = 0 \quad (41)$$

$$\psi_n^{eff,p} = [E_{n+1}^{eff}]^{-\frac{1}{2}} \mathcal{A} \psi_{n+1}^{eff} \quad (42)$$

$$\psi_{n+1}^{eff} = [E_n^{eff,1}]^{-\frac{1}{2}} \mathcal{A}^\dagger \psi_n^{eff,1} \quad (43)$$

The superpotential, corresponding to the potentials derived in case 1 and case 2 in the previous section, can be expressed as

$$\mathcal{W}^I = \frac{b}{2} \left((1 + \alpha) e^{\frac{bx}{2}} - e^{-\frac{bx}{2}} \left(1 + 2 \left(\frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})} - \frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha}(-e^{-bx})} \right) \right) \right) \quad (44)$$

and

$$\mathcal{W}^{II} = \left(\frac{1}{2} - \left(\frac{L_{m-1}^{\alpha+1}(-x^l)}{L_m^{\alpha}(-x^l)} - \frac{L_{m-1}^{\alpha}(-x^l)}{L_m^{\alpha-1}(-x^l)} \right) \right) x^{\frac{l}{2}} + \frac{1 - l(\alpha + 1)}{2l} x^{-\frac{l}{2}} \quad (45)$$

respectively, where $l = \frac{2-\nu}{1-\nu}$.

From the superpotential $\mathcal{W}^I$ (for case 1), the partner potentials can be derived using equations (38) and (39)

$$\begin{aligned} V_{eff}^I = & \frac{b^2}{4} (e^{bx} (\alpha^2 - 1) + e^{-bx}) + b^2 \left(\frac{2L_{m-1}^{\alpha}(-e^{-bx})^2}{(L_m^{\alpha-1}(-e^{-bx}))^2} - \frac{L_{m-2}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})} + \frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{\alpha L_m^{\alpha-1}(-e^{-bx})} \right. \\ & \left. + \frac{L_{m-1}^{\alpha}(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})} - \frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})} \right) e^{-bx} - b^2 \left(\frac{2m + \alpha + \alpha^2}{2\alpha} \right) \end{aligned} \quad (46)$$

$$\begin{aligned} V_{eff}^{I,p} = & \frac{b^2}{4} (e^{bx} \alpha (\alpha + 2) + e^{-bx}) + b^2 \left(\frac{2L_{m-1}^{\alpha+1}(-e^{-bx})^2}{(L_m^{\alpha}(-e^{-bx}))^2} - \frac{L_{m-2}^{\alpha+2}(-e^{-bx})}{L_m^{\alpha}(-e^{-bx})} + \frac{L_{m-1}^{\alpha+2}(-e^{-bx})}{\alpha L_m^{\alpha}(-e^{-bx})} \right. \\ & \left. + \frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha}(-e^{-bx})} - \frac{L_{m-1}^{\alpha+2}(-e^{-bx})}{L_m^{\alpha}(-e^{-bx})} \right) e^{-bx} - b^2 \left(\frac{\alpha + 1}{2} + \frac{m}{\alpha + 2} \right) \end{aligned} \quad (47)$$

Also from superpotential $\mathcal{W}^{II}$ (for case 2) and equations (38) and (39) we can write the partner potentials

$$\begin{aligned} V_{eff}^{II} = & \frac{1}{4} \left(x^{\frac{\nu-2}{1-\nu}} (\alpha^2 - 1) + x^{\frac{2-\nu}{1-\nu}} \right) + \left(\frac{2L_{m-1}^{\alpha}(-x^{\frac{2-\nu}{1-\nu}})^2}{(L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}}))^2} - \frac{L_{m-2}^{\alpha+1}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})} + \frac{L_{m-1}^{\alpha+1}(-x^{\frac{2-\nu}{1-\nu}})}{\alpha L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})} \right. \\ & \left. + \frac{L_{m-1}^{\alpha}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})} - \frac{L_{m-1}^{\alpha+1}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})} \right) x^{\frac{2-\nu}{1-\nu}} + \frac{(1-\nu)(3-\nu)}{4(2-\nu)^2} x^{\frac{\nu-2}{1-\nu}} - \frac{2m + \alpha + \alpha^2}{2\alpha} \end{aligned} \quad (48)$$

and

$$V_{eff}^{II,p} = \frac{1}{4} \left(x^{\frac{\nu-2}{1-\nu}} \alpha(\alpha+2) + x^{\frac{2-\nu}{1-\nu}} \right) + \left(\frac{2L_{m-1}^{(\alpha+1)}(-x^{\frac{2-\nu}{1-\nu}})^2}{(L_m^\alpha(-x^{\frac{2-\nu}{1-\nu}}))^2} - \frac{L_{m-2}^{(\alpha+2)}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^\alpha(-x^{\frac{2-\nu}{1-\nu}})} + \frac{L_{m-1}^{(\alpha+2)}(-x^{\frac{2-\nu}{1-\nu}})}{(\alpha+1)L_m^\alpha(-x^{\frac{2-\nu}{1-\nu}})} \right. \\ \left. + \frac{L_{m-1}^{(\alpha+1)}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^\alpha(-x^{\frac{2-\nu}{1-\nu}})} - \frac{L_{m-1}^{(\alpha+2)}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^\alpha(-x^{\frac{2-\nu}{1-\nu}})} \right) x^{\frac{2-\nu}{1-\nu}} + \frac{(1-\nu)(3-\nu)}{4(2-\nu)^2} x^{\frac{\nu-2}{1-\nu}} - \left(\frac{\alpha}{2} + \frac{m}{\alpha+1} \right) \quad (49)$$

The potentials V_{eff}^I and V_{eff}^{II} obtained in Eq.(46) and (48) are same with potentials in Eq. (23) and (32) for $V_c^I = -b^2 \left(\frac{2m+\alpha+\alpha^2}{2\alpha} \right)$ and $V_c^{II} = -\frac{2m+\alpha+\alpha^2}{2\alpha}$ respectively.

From Eqs. (46) and (47), we note that the potential and its supersymmetric counterpart satisfy the condition

$$V_{eff}^{I,p}(x, \alpha) = V_{eff}^I(x, \alpha+1) + b^2 \quad (50)$$

Also from Eqs. (48) and (49) we see that the partner potentials satisfy the similar relation

$$V_{eff}^{II,p}(x, \alpha) = V_{eff}^{II}(x, \alpha+1) + 1 \quad (51)$$

It's evident that the potentials obtained for case 1 and case 2 satisfy the condition of shape invariant potentials given in Eq. (40). Thus, we deduce that the newly introduced potentials exhibit shape invariance.

We've deduced the eigenstates for the partner potential associated with potential (23) as

$$\psi_{n,m}^p(x) = \alpha \frac{\exp \left[-\frac{1}{2}((\alpha+2)bx + e^{-bx}) \right]}{L_m^\alpha(-e^{-bx})} \hat{L}_{n+m,m}^{\alpha+1}(e^{-bx}) \quad n = 0, 1, 2, \dots \quad (52)$$

and the eigenstate of the partner potential of Eq. (32) is given by

$$\psi_{n,m}^p(x) \propto \frac{\exp \left(\frac{-x^{\frac{2-\nu}{1-\nu}}}{2} \right)}{\left(L_m^\alpha(-x^{\frac{2-\nu}{1-\nu}}) \right)} x^{\frac{(2+\alpha)+(\alpha+1)(1-\nu)}{2(1-\nu)}} \hat{L}_{n+m,m}^{\alpha+1}(x^{\frac{2-\nu}{1-\nu}}), \quad n = 0, 1, 2, \dots \quad (53)$$

5 Results and Discussions

The study of systems endowed with position-dependent mass (PDM) is a subject of great interest in many branches of physics due to its utmost relevance in a wide variety of physical situations. However only a few PDM systems are solved exactly for some specific position dependent of Mass terms, i.e. $M(x) \propto [g'(x)]^\mu$ for $\mu = 1, 2$ and -1 . In this article we have considered a new type of position dependent mass term, which is proportional to $(g'(x))^\nu$, $\nu = \frac{2\eta}{2\eta+1}$, with $\eta = 0, 1, 2, \dots$ and have obtained the solutions of position-dependent mass Schrödinger equation in terms of X_m Laguerre polynomials. Furthermore, we have obtained one parameter family of isochronous potentials, which are exactly solvable. The solution corresponding to these potentials are shown to be in terms of X_m Laguerre polynomials. We have shown the SUSY for PDM systems and for this system, we find the one-parameter family of isochronous potentials and its partner potentials. We show the exact solvability of the system is due to the underlying shape invariance of these SUSY potentials.

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Generating QES potentials supporting zero energy normalizable states for an extended class of truncated Calogero Sutherland model

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Abstract:	It is commonly held that quantum states with energy $E = 0$ would belong to the continuum. However, several situations have been reported when a zero-energy state becomes bound subject to certain restrictions on the coupling constants defining the potential. In the present work, we present another evidence of the existence of regular zero-energy normalizable solutions for a system of QES potentials that correspond to rationally extended many-body truncated Calogero-Sutherland model. Our procedure is based upon the potential group approach with an underlying $so(2, 1)$ structure that utilizes the three cases guided by it on profitably carrying out a point canonical transformation. We deal with each case separately by suitably restricting the coupling parameters.
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Generating QES potentials supporting zero energy normalizable states for an extended class of truncated Calogero Sutherland model

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Abstract

It is commonly held that quantum states with energy $E = 0$ would belong to the continuum. However, several situations have been reported when a zero-energy state becomes bound subject to certain restrictions on the coupling constants defining the potential. In the present work, we present another evidence of the existence of regular zero-energy normalizable solutions for a system of QES potentials that correspond to rationally extended many-body truncated Calogero-Sutherland model. Our procedure is based upon the potential group approach with an underlying $so(2,1)$ structure that utilizes the three cases guided by it on profitably carrying out a point canonical transformation. We deal with each case separately by suitably restricting the coupling parameters.

1 Introduction

A major difficulty in pinning down zero-energy quantal states lies in the extraction of corresponding normalizable solutions of the Schrödinger equation. This has motivated a lot of research in the look out for the existence of such zero-energy states with valid asymptotic properties [1–8]. Interestingly, it was noted [9] long ago that zero-energy solutions could exist with the properties of confinement (in case of normalizability) and leakage (if the wavefunction turns out to be non-normalizable) for certain types of interacting systems. Another example is exemplified in the Coulomb problem, where a zero-energy state is non-normalizable and thus considered a part of the continuum. However, such a situation is not without its exceptions. Indeed, there exist cases pertaining to a QES system where the $E = 0$ state becomes bound should we restrict to certain values of the coupling constant [10].

In a broader context, searching for solvable potentials, that possess an exact spectrum with accompanying closed-form solutions for the wavefunctions, has always been in the spotlight of active research (see, for instance, [11–13]). Among different techniques that have been adopted in this regard, the ones based on point canonical transformation [14] coupled with employing group theoretical techniques that

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exploit the structure of the Casimir operator [15–20], have evinced most attention. On the other hand, the limited number of exactly solvable (ES) potentials in quantum mechanics has prompted investigation into the question of the existence of quasi-exactly solvable (QES) [11, 21–30] and conditionally exactly solvable (CES) systems [31–37]. Note that while the former types allow for a partial or complete assessment of the energy spectrum under possible constraints among the potential parameters, the latter ones are typical in the sense that one or more coupling constants present in them need to be tuned to a specific value thus facilitating a valid asymptotic behaviour. Beyond being of academic interest, both QES and CES potentials provide valuable insights in the characterization of physical phenomena. In particular, a connection between QES and spin-boson and spin-spin interacting systems is well known [38], and the dual partnership between CES and the accompanying shape invariant ES potential was pointed out some time ago [39].

In this work, we present a novel evidence of zero-energy normalizable solutions for a class of QES rational potentials which correspond to rationally extended many-body truncated Calogero-Sutherland model (TCS) [40]. Towards such a pursuit, we make use of the algebraic techniques based upon the profitable employment of $\mathfrak{so}(2, 1)$ as a potential algebra for the Schrödinger equation. A distinguishing feature of our approach is that for all the three categories of solutions furnished by the latter we are led to regular normalizable wavefunctions provided certain convergence condition is satisfied. The structure of this paper is organized as follows: Section 2 provides a concise review of the $\mathfrak{so}(2, 1)$ potential algebra discussing its basic layout. In Section 3, we introduce the rationally extended many-body truncated Calogero-Sutherland model (TCS) while in Section 4 we utilize the relevance of $\mathfrak{so}(2, 1)$ to determine the zero-energy normalizable solutions for a specific class of QES rational potentials within TCS, subject to certain restrictions on the potential parameters. Finally, in Section 5, a summary is presented.

2 $\mathfrak{so}(2, 1)$ potential algebra

The underlying commutation relations of $\mathfrak{so}(2, 1)$, namely $[J_+, J_-] = -2J_0$, $[J_0, J_\pm] = \pm J_\pm$, are guided by the following differential realizations of the generators

$$J_0 = -i\frac{\partial}{\partial\phi}, \quad J_\pm = e^{\pm i\phi} \left[\frac{\partial}{\partial x} + F(x) \left(i\frac{\partial}{\partial\phi} \mp \frac{1}{2} \right) + G(x) \right], \quad (1)$$

where the two functions F and G are controlled by a set of coupled equations

$$F' = 1 - F^2, \quad \text{and} \quad G' = -FG, \quad (2)$$

in which the dashes denoting derivatives with respect to x . For bound states, to which we restrict here, one considers unitary irreducible representations of the D_k^+ -type that point to basis states which correspond to the eigenfunctions of different Hamiltonians, but conforming to the same energy level. Note that the Casimir operator being $J^2 = J_0^2 \mp J_0 - J_\pm J_\mp$, it has an explicit form $J^2 = (\partial^2/\partial x^2) - F'[(\partial^2/\partial\phi^2) + \frac{1}{4}] + 2iG'(\partial/\partial\phi) - G^2 - \frac{1}{4}$. Hence, with $J_0|km\rangle = m|km\rangle$, and $J^2|km\rangle = k(k-1)|km\rangle$ ($m = k, k+1, k+2, \dots$), where, in the realization of (1), the states given by $|km\rangle = \psi_{km}(x)e^{im\phi}$ are the basis functions, it readily follows that the coefficient functions $\psi_{km}(x)$ satisfy the Schrödinger equation

$$-\psi_{km}'' + V_m\psi_{km} = -\left(k - \frac{1}{2}\right)^2 \psi_{km} \quad (3)$$

in the presence of a one-parameter family of potentials expressed in terms of $F(x)$ and $G(x)$

$$V_m = \left(\frac{1}{4} - m^2\right) F' + 2mG' + G^2. \quad (4)$$

As shown by Wu and Alhassid [15–17], a particular choice of solutions to (2) correspond to Morse, Pöschl-Teller, and Rosen-Morse potentials. Later, an extended formalism by Englefield and Quesne [18]

identified more general possibilities to include additional solutions according to whether $F^2 < 1$, $F^2 = 1$ or $F^2 > 1$ namely,

$$(I) \quad F(x) = \tanh(x), \quad G(x) = b \operatorname{sech}(x), \quad (5)$$

$$(II) \quad F(x) = \pm 1, \quad G(x) = b e^{\mp x}, \quad (6)$$

$$(III) \quad F(x) = \coth(x), \quad G(x) = b \operatorname{cosech}(x). \quad (7)$$

By substituting these solutions into equation (4) one arrives at non-singular Gendenshtein, Morse, and singular Gendenshtein potentials. These solutions encompass those obtained by Wu and Alhassid in that Gendenshtein potentials are disguised versions of Pöschl-Teller potentials; further, for particular values of the parameters, one gets non-singular Rosen-Morse potentials from singular Gendenshtein potentials.

3 Extended truncated Calogero-Sutherland model

The extended N -body TCS model in one-dimension is concerned with N particles which are harmonically confined and interact pairwise via two-body potential as well in the presence of a three-body term. The idea of truncation of the interaction refers to the number of neighbours instead of the relative distance of the particles. The relevant Hamiltonian is characterized by the form

$$H_{ext} = \hat{H} + V_{new} \quad (8)$$

where following the notations of [40, 41]

$$\hat{H} = \sum_{i=1}^N \left[-\frac{1}{2} \frac{\partial^2}{\partial x_i^2} + \frac{1}{2} \omega^2 x_i^2 \right] + V_{int}, \quad (9)$$

$$V_{int} = \sum_{\substack{i < j \\ |i-j| \leq r}} \frac{\lambda(\lambda-1)}{|x_i - x_j|^2} + \sum_{\substack{i < j < k \\ |i-j| \leq r \\ |j-k| \leq r}} \frac{\lambda^2 \mathbf{r}_{ij} \cdot \mathbf{r}_{jk}}{r_{ij}^2 r_{jk}^2}; \quad \lambda \neq 0, \quad (10)$$

and V_{new} is a new interaction term as given by

$$V_{new} = \frac{(\alpha_1 + \alpha_2 \omega^2 \rho^2)}{(\beta_1 + \beta_2 \omega^2 \rho^2)^2} \quad (11)$$

containing some unknown coefficients. The two body interactions in the above scheme are attractive in the range $0 < \lambda < 1$, and repulsive for $\lambda \geq 1$. It needs to be pointed out that for the specific case of $r = 1$, the corresponding form of the Hamiltonian goes over to the model of Jain and Khare [42], while for the case $r = N - 1$, one recovers what was initially proposed by Calogero [43], and Sutherland [44]. ρ in Eq. 11 varies from 0 to ∞ unlike the variable x in $so(2, 1)$ potential algebra, which varies from $-\infty$ to ∞ .

To find the solution of the Schrödinger equation accompanying Eq. (8)

$$\hat{H}_{ext} \Psi_{ext} = E_{ext} \Psi_{ext} \quad (12)$$

we consider $\Psi_{ext}(\mathbf{x}) = \phi(\mathbf{x}) \xi_{ext}$, where $\phi(\mathbf{x})$ is given by $\prod_{i < j} (x_i - x_j)^\lambda$, and ξ_{ext} satisfying

$$-\frac{1}{2} \sum_{i=1}^N \frac{\partial^2 \xi_{ext}}{\partial x_i^2} - \lambda \sum_{i < j}^{N-1} \frac{1}{x_i - x_j} \left(\frac{\partial \xi_{ext}}{\partial x_i} - \frac{\partial \xi_{ext}}{\partial x_j} \right) + \left(\frac{1}{2} \sum_{i=1}^N \omega^2 x_i^2 + V_{new} - E_{ext} \right) \xi_{ext} = 0. \quad (13)$$

Next, redefining the function $\xi_{ext} = \Phi_{ext}(\rho)P_s(\mathbf{x})$, we find that $\Phi_{ext}(\rho)$ obeys the radial equation [41]

$$\Phi_{ext}''(\rho) + (N + 2s - 1 + \lambda r(2N - r - 1))\frac{1}{\rho}\Phi_{ext}'(\rho) + 2(E - (\frac{1}{2}\omega^2\rho^2 + V_{new}))\Phi_{ext}(\rho) = 0 \quad (14)$$

with $P_s(\mathbf{x})$ given by the Laplace equation. Note that in (14) a prime on $\Phi_{ext}(\rho)$ indicates derivative with respect to ρ . To solve it we follow the standard procedure [45] of substituting

$$\Phi_{ext}(\rho) = f(\rho)\zeta(g(\rho)), \quad (15)$$

where $f(\rho)$ is some unknown coefficient function to be determined, $g(\rho)$ is a variable transformation and $\zeta(g)$ is to be so selected from the consideration of its fulfilling a second-order differential equation

$$\zeta''(g(\rho)) + Q_1(g)\zeta'(g(\rho)) + R_1(g)\zeta(g(\rho)) = 0. \quad (16)$$

such that the functions $Q_1(g)$ and $R_1(g)$ are well defined if a special function choice is made for it. Inserting Eq. (15) into Eq. (14), we get

$$\zeta''(g) + \left(\frac{2f'(\rho)}{f(\rho)g'(\rho)} + \frac{g''(\rho)}{g'(\rho)^2} + \frac{\tau}{\rho g'(\rho)} \right) \zeta'(g) + \frac{1}{g'(\rho)^2} \left(\frac{f''(\rho)}{f(\rho)} + \frac{\tau f'(\rho)}{\rho f(\rho)} + 2(E_{ext} - V_{ext}) \right) \zeta(g) = 0, \quad (17)$$

where $V_{ext} = \frac{1}{2}\omega\rho^2 + V_{new}$ and $\tau = (N + 2s - 1 + \lambda r(2N - r - 1))$. On comparing Eq. (17) with Eq. (16), we get

$$Q_1(g) = \frac{2f'(\rho)}{f(\rho)g'(\rho)} + \frac{g''(\rho)}{g'(\rho)^2} + \frac{\tau}{\rho g'(\rho)} \quad \text{and} \quad R_1(g) = \frac{1}{g'(\rho)^2} \left(\frac{f''(\rho)}{f(\rho)} + \frac{\tau f'(\rho)}{\rho f(\rho)} + 2(E_{ext} - V_{ext}) \right). \quad (18)$$

After simplifying $Q_1(g)$, one finds that

$$f(\rho) \simeq (g'(\rho))^{-\frac{1}{2}} \rho^{-\frac{\tau}{2}} \exp \left(\frac{1}{2} \int^g Q_1(g) dg \right). \quad (19)$$

Using it in the expression of $R_1(g)$ we arrive at the form

$$E_{ext} - V_{ext} = \frac{1}{2} \left[\frac{g'''(\rho)}{2g'(\rho)} - \frac{3}{4} \frac{g''(\rho)^2}{g(\rho)^2} + \frac{\tau/2(\tau/2 - 1)}{\rho^2} + g'(\rho)^2 \left(R_1(g) - \frac{Q_1'(g)}{2} - \frac{Q_1^2(g)}{4} \right) \right]. \quad (20)$$

which can be further simplified to

$$E_{ext} - V_{ext} = \frac{1}{2} \left[\frac{\Delta V(\rho)}{2} + \frac{\tau/2(\tau/2 - 1)}{\rho^2} + g'(\rho)^2 \left(R_1(g) - \frac{Q_1'(g)}{2} - \frac{Q_1^2(g)}{4} \right) \right]. \quad (21)$$

where the quantity $\Delta V(\rho)$ is the so-called Schwartzian derivative,

$$\Delta V(\rho) = \frac{g'''(\rho)}{g'(\rho)} - \frac{3}{2} \frac{g''(\rho)^2}{g(\rho)^2}. \quad (22)$$

(21) is the main relation which we will exploit below in search for normalizable zero-energy solutions that reside in an extended class of truncated Calogero Sutherland model.

4 Normalizable zero-energy solutions

To find QES systems with normalizable zero-energy solutions we follow the procedure in [10]. Towards this end, we first of all set $Q_1(g) = 0$ which gives the condition

$$f(\rho)^2 g'(\rho) \rho^\tau = \text{constant} \quad (23)$$

This reduces equation (21) to the form

$$E_{ext} - V_{ext} = \frac{\Delta V(\rho)}{4} + \frac{\tau/4(\tau/2 - 1)}{\rho^2} + \frac{1}{2}g'(\rho)^2 R_1(g). \quad (24)$$

Next, to get a meaningful representation of (24) we put $R_1 = E_T - V_T(g)$ and use the transformation $\rho = K(g(\rho))$ to get

$$E_T - V_T(g) = [E_{ext} - V_{ext}]2K'(\rho)^2 + \frac{1}{2}\Delta V(K(g)) - \frac{\frac{\tau}{2}(\frac{\tau}{2} - 1)}{K(g)^2}K'(g)^2 \quad (25)$$

Now using $so(2, 1)$ as potential algebra of the Schrodinger equation and putting $V_T = V_m$

$$\left(\frac{1}{4} - m^2\right)(1 - F^2) - 2mFG + G^2 - E_T = 2[K'(g)]^2[V_{ext}(K(g)) - E_{ext}] - \frac{1}{2}\Delta V(K(g)) + \frac{\frac{\tau}{2}(\frac{\tau}{2} - 1)}{K(g)^2}K'(g)^2, \quad (26)$$

Now taking first class of solution (5), using mapping function $\rho = K(g) = (e^g - 1)^{-1}$ and setting $E = 0$ gives

$$\begin{aligned} V(K(g)) &= (1 - 4m^2 + 4b^2) \operatorname{sech} g (\operatorname{sech} g - 2) - 8mb(\tanh g \operatorname{sech} g - 2 \tanh g + \sinh g) \\ &\quad - \left(4E_T + \frac{1}{2}\right) \cosh g (\cosh g - 2) - 4 \left(E_T + m^2 - b^2 - \frac{1}{8}\right) + \frac{\tau}{4} \left(\frac{\tau}{2} - 1\right) (2 \sinh^2 g + \sinh g) \end{aligned}$$

Thus by using the mapping function and above equation we can find $V(\rho)$

$$(I) \quad V(\rho) = -\frac{A}{(2\rho^2 + 2\rho + 1)^2} - B \frac{2\rho + 1}{\rho(\rho + 1)(2\rho^2 + 2\rho + 1)^2} + \frac{C}{\rho^2(\rho + 1)^2} - \frac{\frac{\tau}{4}(\frac{\tau}{2} - 1)}{\rho^2}, \quad (27)$$

with

$$A = \frac{1}{2}(4(m^2 - b^2) - 1), \quad B = 2mb, \quad C = -\frac{1}{2}\left(E_T + \frac{1}{4}\right) = \left(k - \frac{1}{2}\right)^2 - \frac{1}{4} = k(k - 1), \quad (28)$$

The first example of a QES potential with known $E = 0$ is given in equation (27). For nonnegative values of C and $\tau = 0$, its behaviour at the origin is similar to that of the (shifted) Coulomb effective potential $V(\rho) = \frac{\hbar^2}{2m} \frac{l(l+1)}{\rho^2} - \frac{e^2}{\rho}$. The corresponding wave function for the first class potential with $m = k$ is $\psi(\rho) = \sqrt{K'(g(\rho))}\chi_0$, where $\chi_0 \sim G^{k-\frac{1}{2}}h$ is the ground state wave function of class I potential in Eq. (5) with $h = \exp[b \tan^{-1}(\sinh g)]$ [18] which gives

$$\psi(\rho) = 2^{k-\frac{1}{2}} \sqrt{\rho(\rho + 1)} \rho^{-\frac{\tau}{2}} \left(\frac{b\rho(\rho + 1)}{2\rho(\rho + 1) + 1} \right)^{k-\frac{1}{2}} e^{b \tan^{-1}\left(\frac{2\rho+1}{2\rho^2+2\rho}\right)} \quad (29)$$

From the above expression, we can clearly see that for $\rho \rightarrow \infty$, the expression depends only on $\rho^{1-\frac{\tau}{2}}$. For the wavefunction to be well-behaved, the power must be negative. This imposes a constraint that $\tau > 2$. On the other hand, for $\rho \rightarrow 0$, the expression depends on $\rho^{k-\frac{\tau}{2}}$. For the wavefunction to be well-behaved in this case, the power $k - \frac{\tau}{2}$ must be positive, which imposes a constraint on k such that $2k > \tau$. In summary, the wavefunction is clearly well-behaved provided we put on restrictions $2k > \tau$ and $\tau > 2$. Figure 1 shows the plots of the probability density and wave function for different values of k , τ , and b .

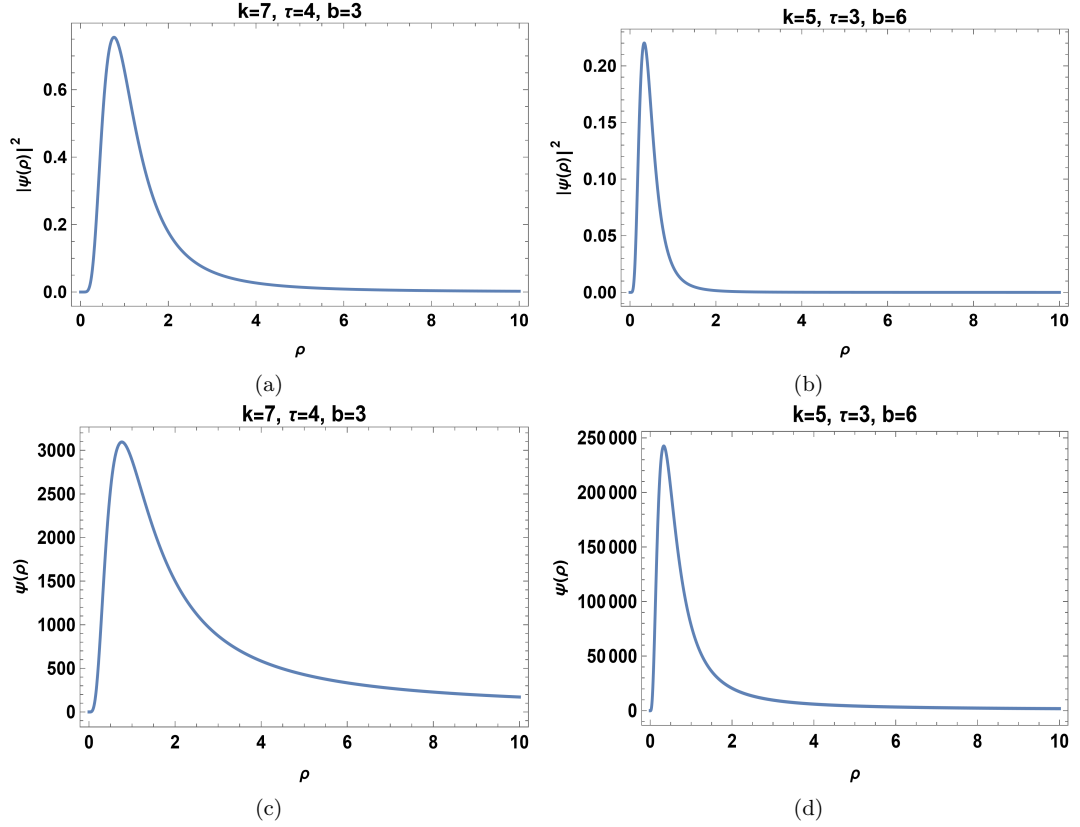


Figure 1: (a) & (b) The plot illustrates the probability density for the first class potential at different values of k , τ , and b (c) & (d) The plot illustrates the wavefunction for the first class potential at different values of k , τ , and b

Similarly we can find the QES potential for other two class.

$$(II) \quad V(\rho) = \frac{b^2}{2(\rho+1)^4} - \frac{mb}{\rho(\rho+1)^3} - \frac{E_T + \frac{1}{4}}{2\rho^2(\rho+1)^2} - \frac{\frac{\tau}{4}(\frac{\tau}{2} - 1)}{\rho^2} \quad (30)$$

The zero energy wave function for the second class potential with $m = k$ is $\psi(\rho) = \sqrt{K'(g(\rho))}\chi_0$, where $\chi_0 \sim G^{k-\frac{1}{2}}h$ with $h = \exp[-be^{-g}]$ which gives

$$\psi(\rho) = \sqrt{\rho(\rho+1)}e^{-\frac{b\rho}{\rho+1}}\rho^{-\frac{\tau}{2}}\left(\frac{b\rho}{\rho+1}\right)^{k-\frac{1}{2}} \quad (31)$$

It is clear that $\psi(\rho)$ has an acceptable asymptotic behaviour provided the conditions $2k > \tau$, $\tau \geq 4$, and $b > 0$ are met. A graphical illustration of the probability density and the wavefunction is shown in Figure 2 for different values of k , τ , and b .

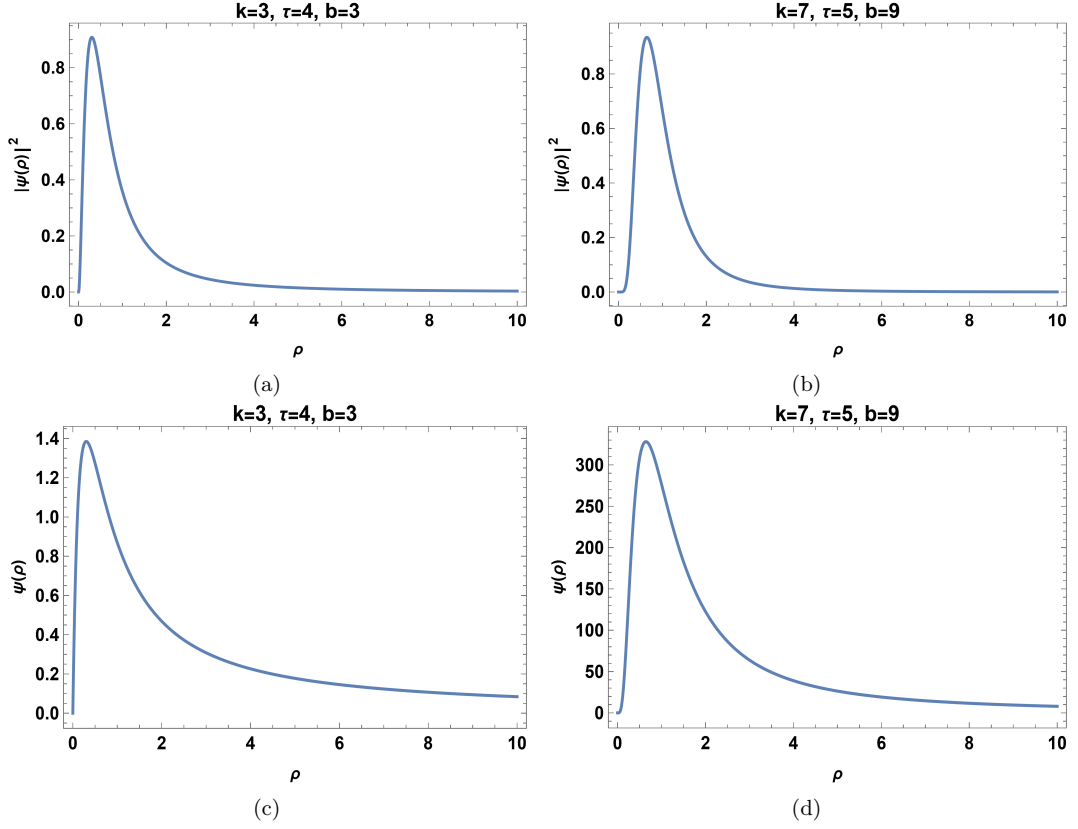


Figure 2: (a) & (b) The plot illustrates the probability density for the second class potential at different values of k , τ , and b (c) & (d) The plot illustrates the wavefunction for the second class potential at different values of k , τ , and b

The third kind of QES potential has the form

$$(III) \quad V(\rho) = \frac{4(m+b)^2 - 1}{2(2\rho+1)^2} - \frac{2mb}{\rho(\rho+1)} - \frac{E_T + \frac{1}{4}}{2\rho^2(\rho+1)^2} - \frac{\frac{\tau}{4}(\frac{\tau}{2} - 1)}{\rho^2} \quad (32)$$

Its supporting wave function with $m = k$ is given by

$$\psi(\rho) = 2^{k-\frac{1}{2}} \sqrt{\rho(\rho+1)} \left(\frac{1}{2\rho+1} \right)^b \rho^{-\frac{\tau}{2}} \left(\frac{b\rho(\rho+1)}{2\rho+1} \right)^{k-\frac{1}{2}} \quad (33)$$

The normalization integral is given by

$$\int_0^\infty |\psi|^2 d\rho = \int_0^\infty \frac{\rho^{-\tau} [\rho(\rho+1)]^{2k}}{(2\rho+1)^{2k+2b-1}} d\rho \quad (34)$$

To examine its convergence, we apply the criterion which states that if $\lim_{\rho \rightarrow \infty} \rho^\alpha f(\rho) = L \neq 0$ and $\alpha \leq 1$, then $\int_a^\infty f(\rho) d\rho$ diverges. In this case, as $\rho \rightarrow \infty$, $f(\rho) \sim \rho^{2k-2b-\tau+1}$. Setting $\alpha = 2b-2k+\tau-1$, we find that $\lim_{\rho \rightarrow \infty} \rho^\alpha f(\rho) = L \neq 0$. The integral converges if $2b-2k+\tau-1 > 1$ or $b > k - \frac{\tau}{2} + 1$, and diverges if $b \leq k - \frac{\tau}{2} + 1$.

It can be seen to be well-behaved when the conditions $2k > \tau$ and $b > k - \frac{\tau}{2} + 1$ hold. The plots of the probability density and the wavefunction are shown in Figure 3 for different values of k , τ , and b . Representative graphs of the potential for different choices of the parameters are sketched in Figure 4.

5 Summary

To summarize, we have dealt with three cases corresponding to the restrictions on the guiding functions of $\text{so}(2, 1)$. An important outcome of our analysis is that in all the three cases we found normalizable wavefunctions by placing plausible restrictions on the coupling parameters. This is in marked contrast to a similar situation one faced in a previous work [10] while addressing a class of QES rational potentials with known $E = 0$ eigenvalue. It was found that subject to certain convergence condition being satisfied, the wave function of the zero-energy state for only one class of potentials turned out to be normalizable. Addressing a rationally extended truncated Calogero-Sutherland model, we subjected the associated Schrödinger equation to a point canonical transformation and matched the resulting energy expression with the Casimir of the $\text{so}(2, 1)$ algebra. This gave three new classes of QES rational potentials corresponding to the $E = 0$ state. For all of them, we showed that the solutions for the eigenvalue equation at zero energy were normalizable and asymptotically smooth provided the coupling parameters were properly restricted.

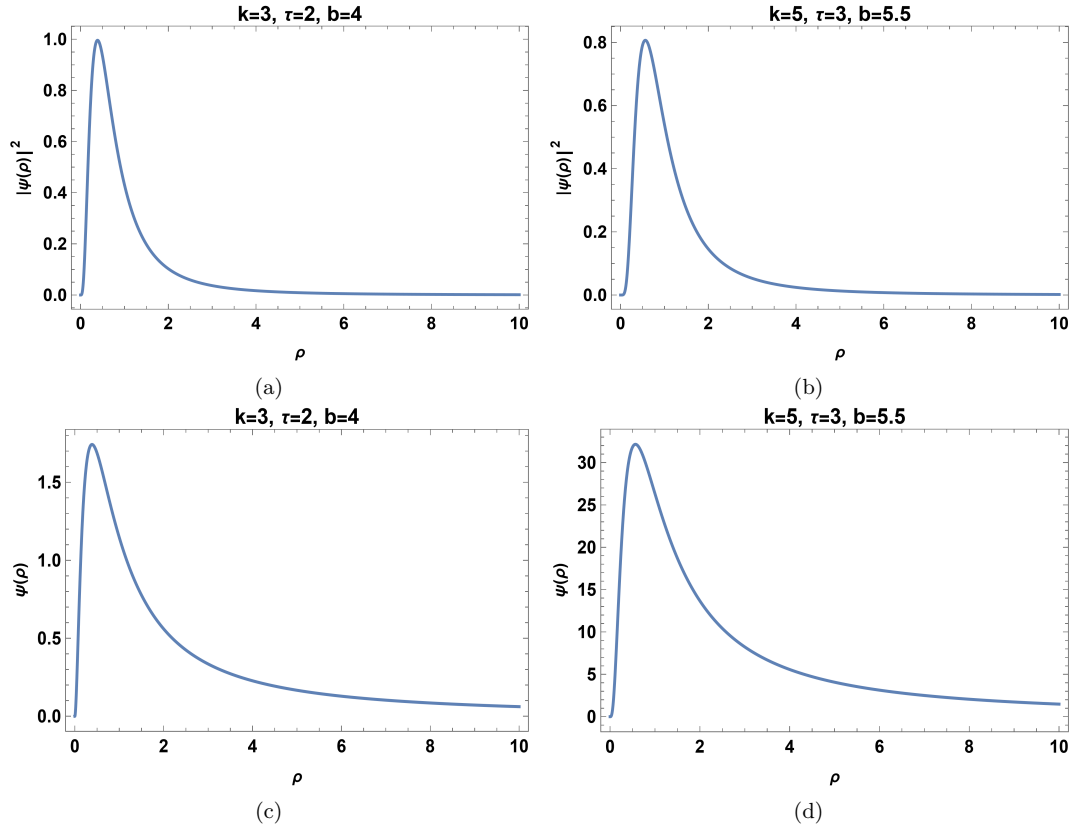


Figure 3: (a) & (b) The plot illustrates the probability density for the third class potential at different values of k , τ , and b (c) & (d) The plot illustrates the wavefunction for the third class potential at different values of k , τ , and b

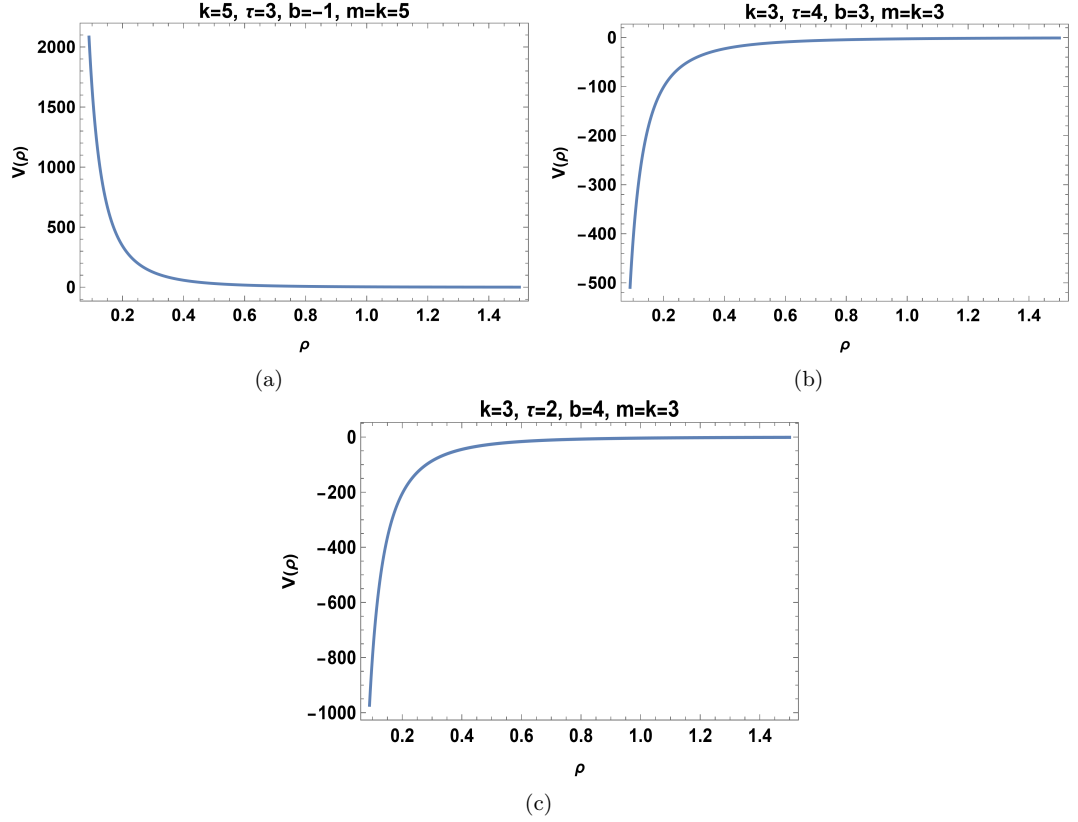


Figure 4: (a) The plots of potential $V(\rho)$ of equation (27). (b) The plot of potential $V(\rho)$ of equation (30). (c) The plot of potential $V(\rho)$ of equation (32).

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7 Data availability statement

All data supporting the findings of this study are included in the article.

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